# Assignment 3: Complete Data Analysis

## Statement of Contribution

| Deliverable | Sebastian | Gary | Lenard |
|---|---|---|---|
| Data Preprocessing | All | All | All |
| Generating t-SNE plots | All | All | All |
| Hierarchical clustering | All | All | All |
| Feature Selection | All | All | All |
| Classification Learner App (Machine Learning) | All | All | All |
| SPSS Question 4 | All | All | All |
| SPSS Question 5 | All | All | All |
| SPSS Question 6 | All | All | All |
| Debugging | All | All | All |
| Report Writing | All | All | All |

Group members had meetings to plan and discuss findings for each component of the assignment. Members shared the code with each other, taking turns writing and debugging. Moreover, each group member had an equal and significant contribution in writing the final report.

By submitting this assignment all group members agree to the above listed contributions to be true and accurate.

## Initial Gene Expression and IQR Visualization on Raw Scale

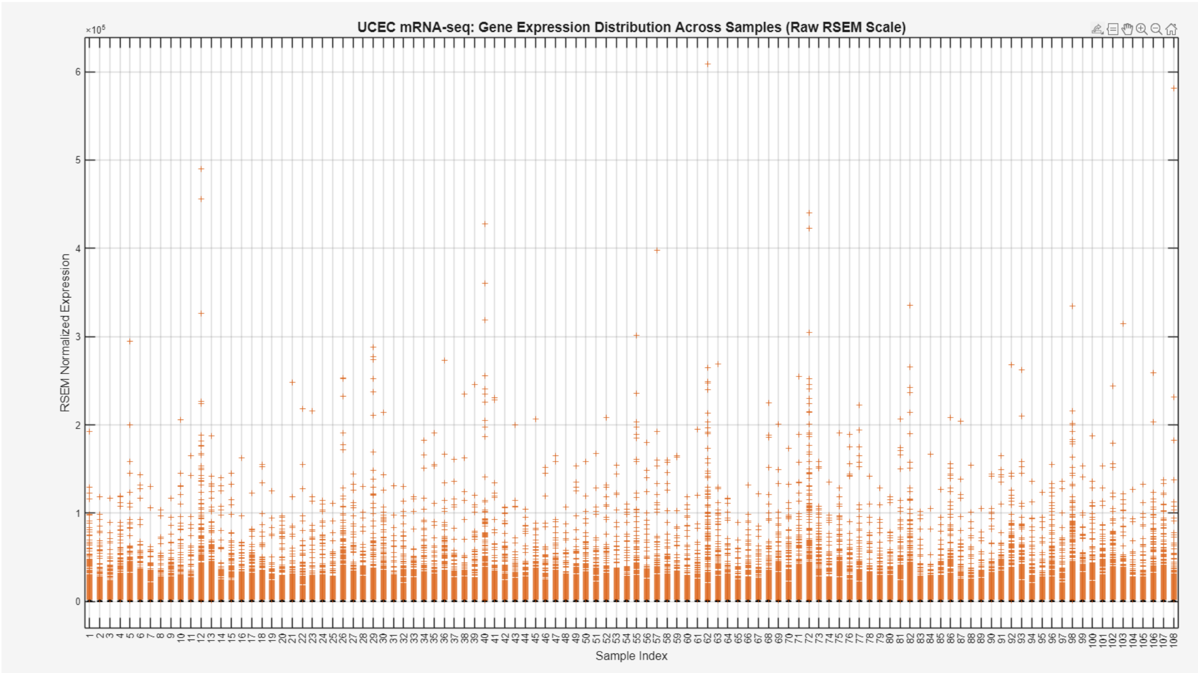


*Figure 1.* Initial Gene Expression and Distribution Across Raw Samples for UCEC mRNA-seq Sample

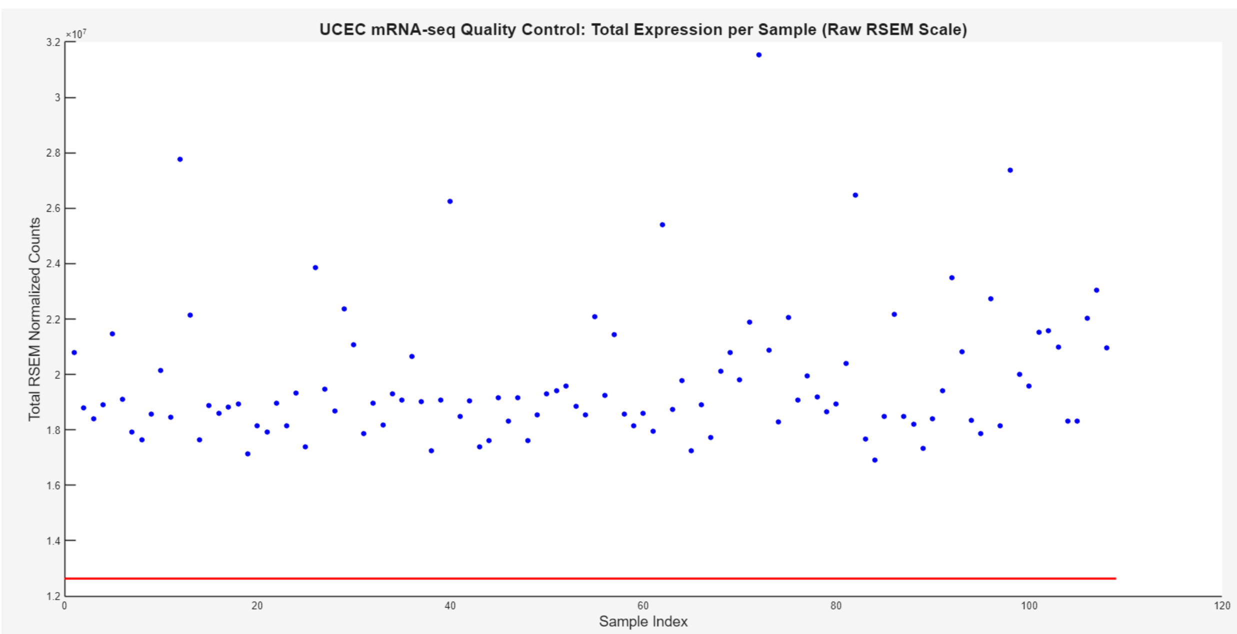


*Figure 2.* Plot of total expression per sample

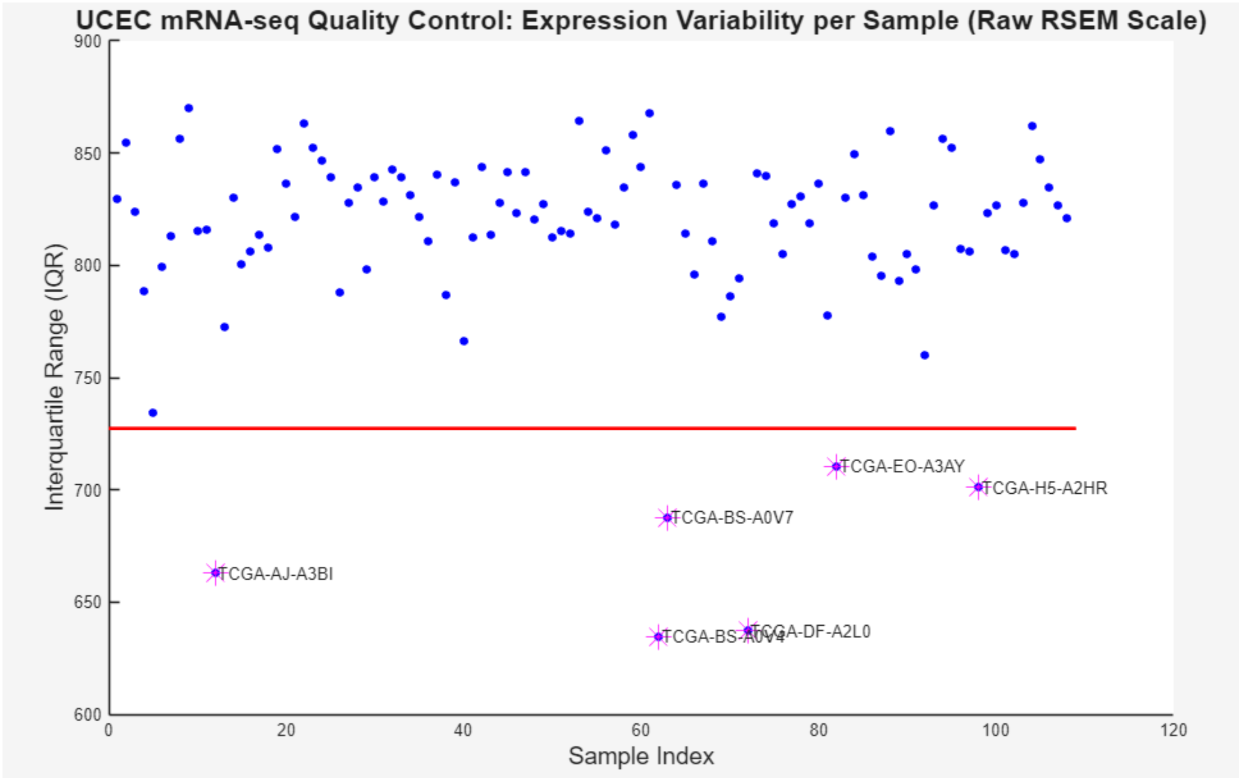


*Figure 3.* Expression variability per sample in the UCEC mRNA-seq data set

*Figure 1* illustrates the raw data for mRNA expression for each sample is heavily skewed, with the inability to clearly see the median and interquartile range for each sample. While the total counts of each sample is above the alpha threshold for outlier detection as seen in *Figure 2*, Figure 3 illustrates the interquartile range of raw RSEM mRNA expressions shown for each sample, used as a quality control metric to identify samples that have low variability. The horizontal red line at 730 indicates the lower IQR threshold to flag potential outliers. The 6 samples that fall below this threshold are TCGA-AJ-AJ-A3BI, TCGA-BS-A0V7, TCGA-BS0V4, TCCA-DF-A2L0, TCGA-EO-A3AY, and TCGA-H5-A2HR. Nonetheless, considering that the raw data is skewed, log2 transformation will be conducted first and then quality control will be applied to the dataset.

**Gene Expression and IQR Visualization Log-Transformed**

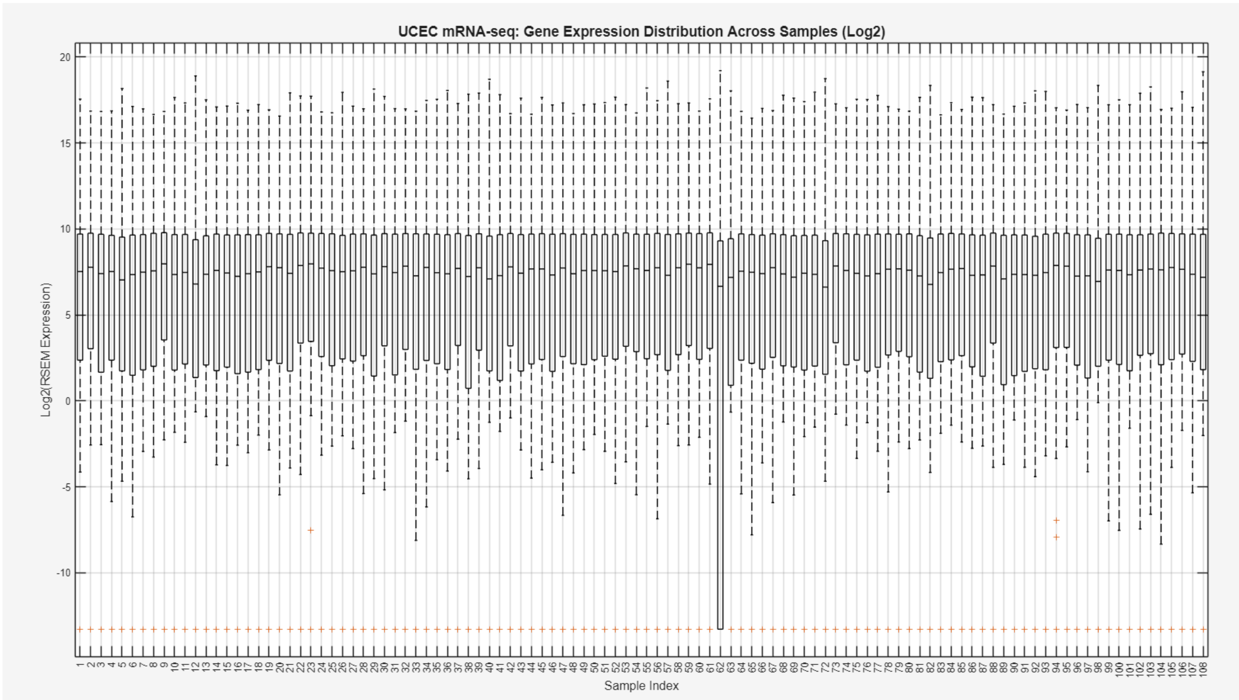


*Figure 4.* Log2 Transformed mRNA expression distributions across UCEC

The box plots in figure 4 show the distribution of the log2 transformed mRNA values for each UCEC sample. The majority of the samples display similar median expression levels indicating consistent expression distributions after transformation. There is a small number of samples that have reduced spread consistent with the low variability samples that were found in figure 3. Sample 62 in particular illustrates data that might be skewed, as the lower quartile extends much further from that seen in the rest of the samples. This will be investigated further.

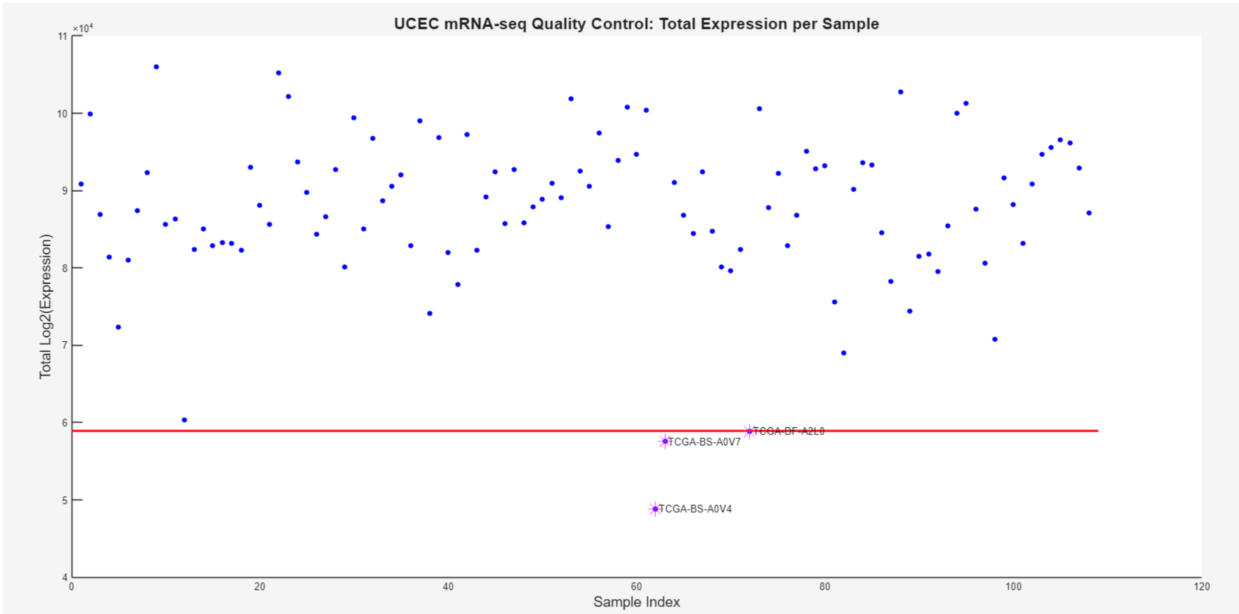


*Figure 5.* Total mRNA expression per sample in the UEC mRNA-seq dataset

The total mRNA expression is shown for each sample, in figure 5, the horizontal red line is the lower threshold to identify potential outliers. While analyzing total expression reveals three outliers (TCGA-BS-A0V4, TCGA-BS-A0V7, and TCGA-DF-A2L0), further exploration of outliers using IQR and average Pearson correlation will help to identify if these identified outliers should be kept or removed.

**Outlier Removal for IQR and Spearman Correlation**

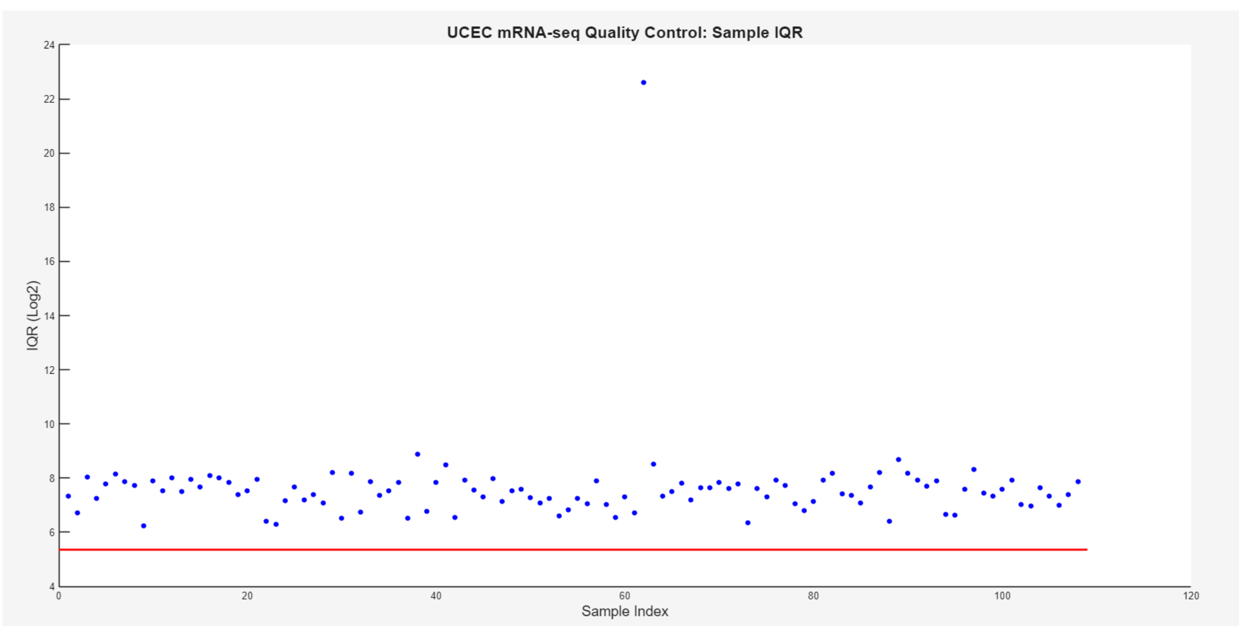


*Figure 6. Performing quality control by investigating IQR of each sample*

When investigating the IQR of each sample as seen in Figure 6, all samples had an IQR above the alpha threshold for detecting outliers. Moreover, no samples indicated technical failure where IQR is equal to 0. Nonetheless, Figure 7 reveals 8 potential outliers that are below the alpha threshold. These outliers include samples TCGA-AX-A05Y, TCGA-AX-A2HC, TCGA-AX-A3FV, TCGA-AX-A3FZ, TCGA-BG-A3EW, TCGA-BK-A4ZD, TCGA-BS-A0V4, and TCGA-BS-A0V7. While these samples lie below the threshold, it cannot be excluded right away and must be investigated considering that the samples do not lie further than a correlation of 0.2 from other samples.

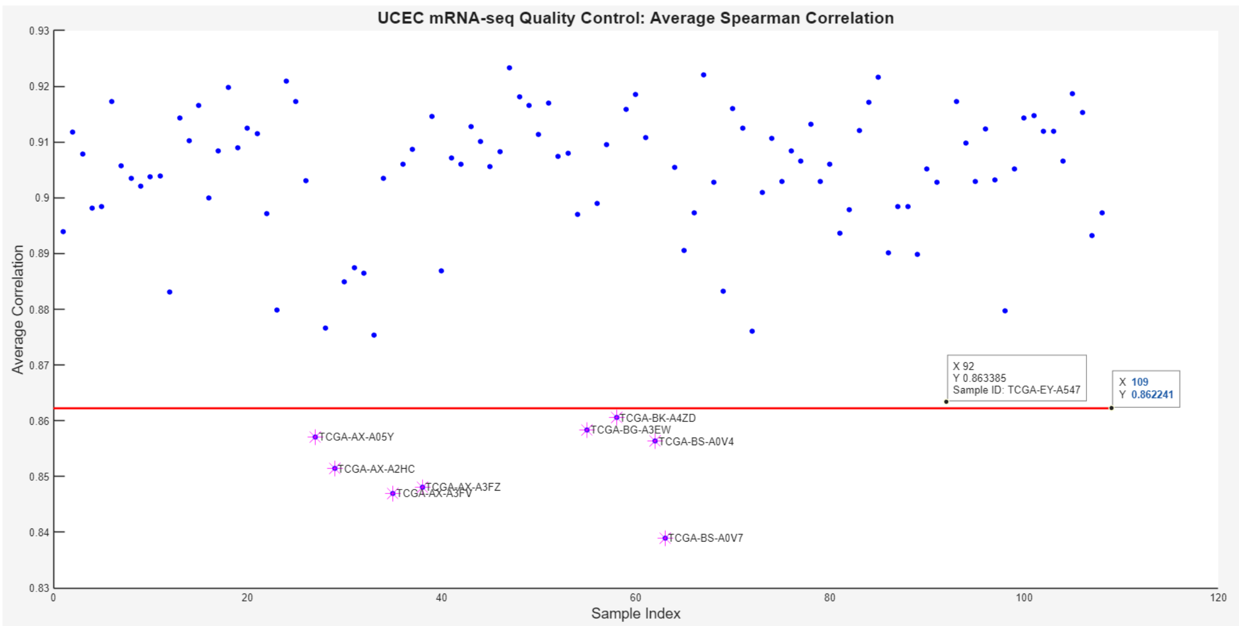


*Figure 7. Performing quality control by investigating average spearman correlation of each sample*

## <u>Post-QC Visualization</u>

Below, figures 8, 9, and 10 illustrate a comparison of IQR, average Pearson correlation, and boxplots of samples after the removal of the 8 outliers previously identified.

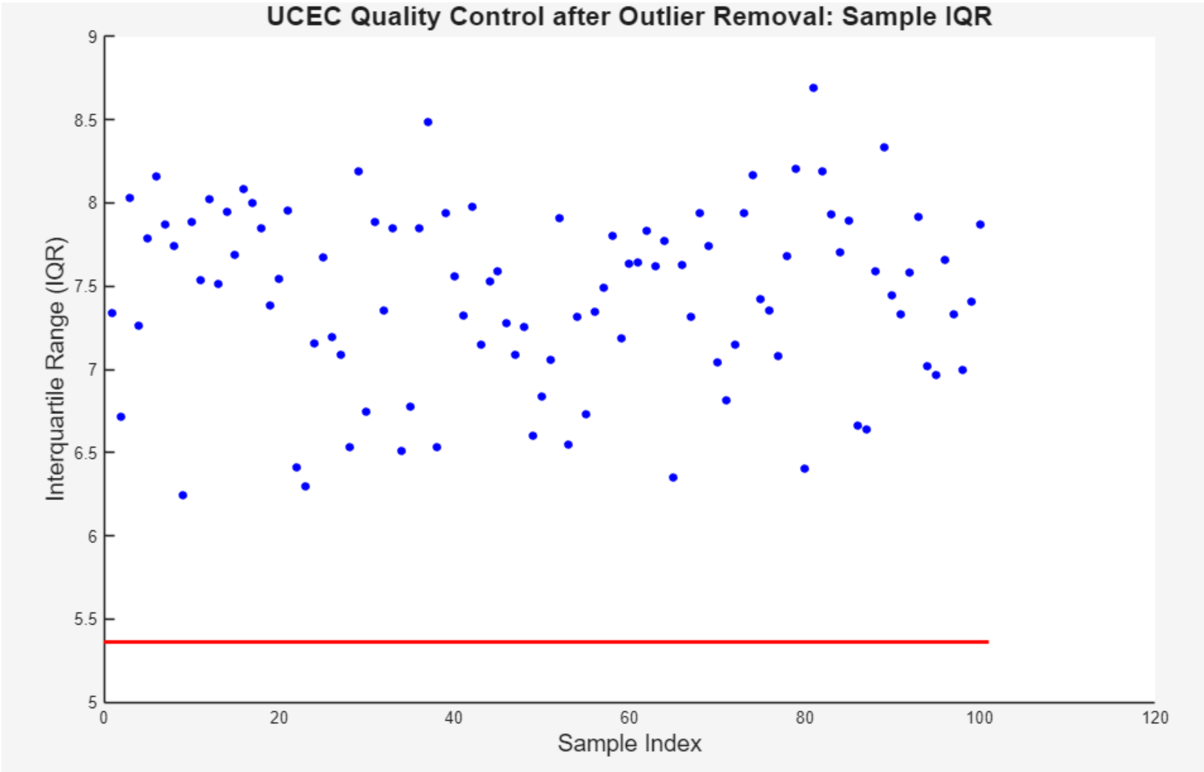


*Figure 8. IQR of samples after removal of 8 outliers identified in average Pearson correlation plot*

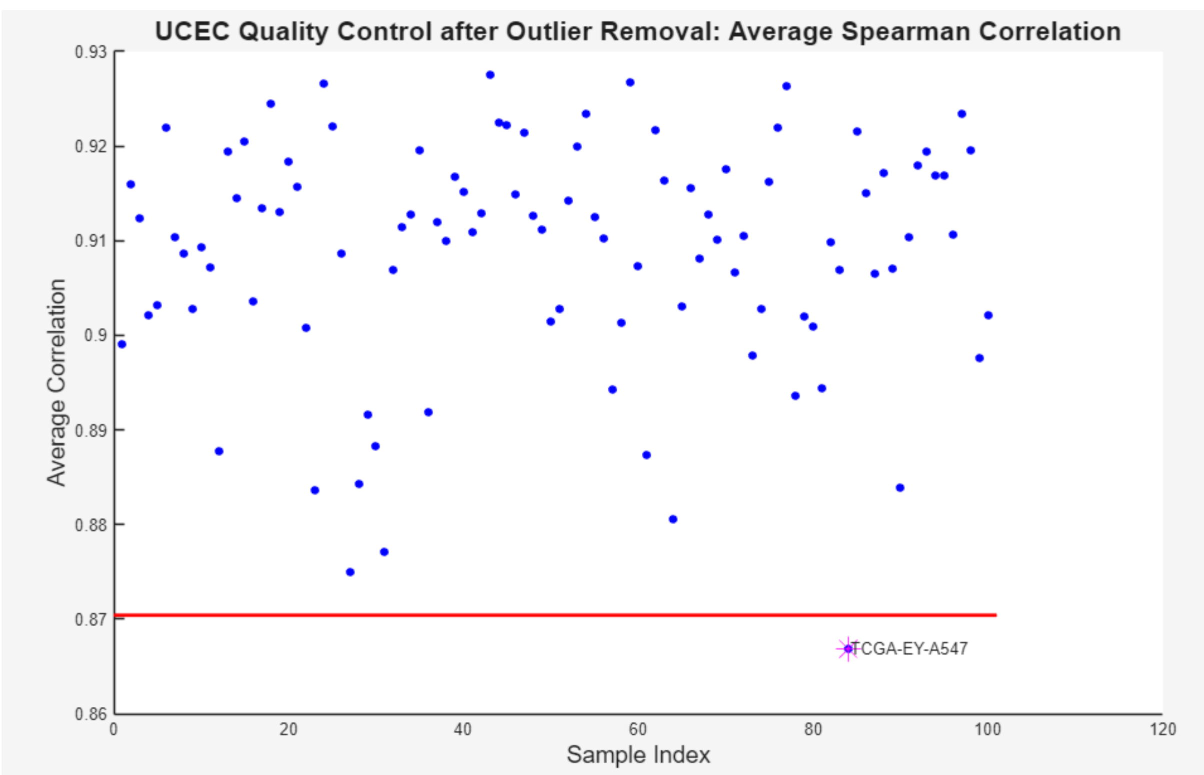


*Figure 9. Average Pearson correlation of samples after removal of 8 outliers identified*

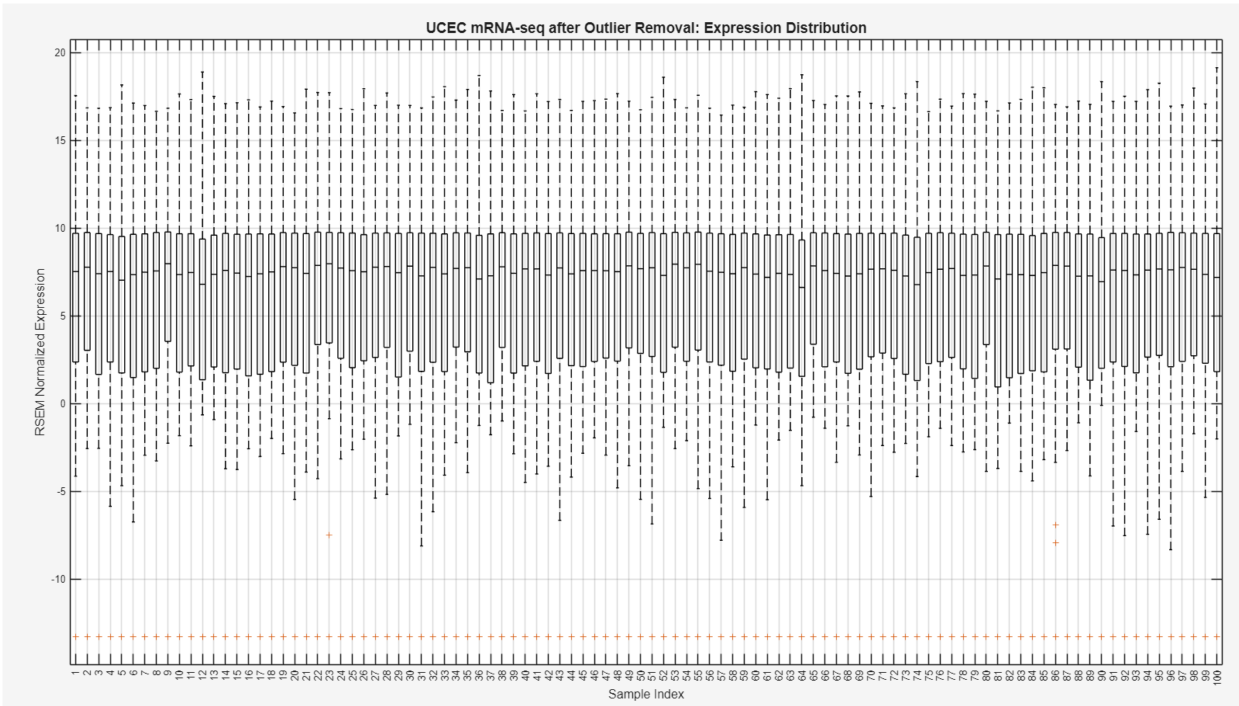


*Figure 10. Boxplot demonstrating alignment after removal of 8 outliers*

As illustrated in Figure 8, the removal of the 8 outliers did not result in any samples being below the alpha threshold. In Figure 9, the average correlation of samples in comparison to all other samples changed when the 8 outliers were removed. Before outlier removal, average Pearson correlation ranged from 0.87 to 0.92. After outlier removal, average Pearson correlation ranged from 0.87 to 0.93. While there is only a slight change in average Pearson correlation of the samples, Figure 10 illustrating a boxplot of expression distribution for all samples, shows a more consistent distribution across the median. There is better alignment of the boxes in comparison to before outlier removal. For example, sample 62 was removed after outlier removal.

One outlier was observed to be below the alpha threshold in Figure 9. Nonetheless, considering its close average Pearson correlation to all other samples (approximately 0.87) and the good alignment with samples in the boxplot seen in *Figure 10,* this sample was not removed from the dataset.

**Final Data Visualization after 0.90 Threshold Filtering with 8 Outliers Removed**

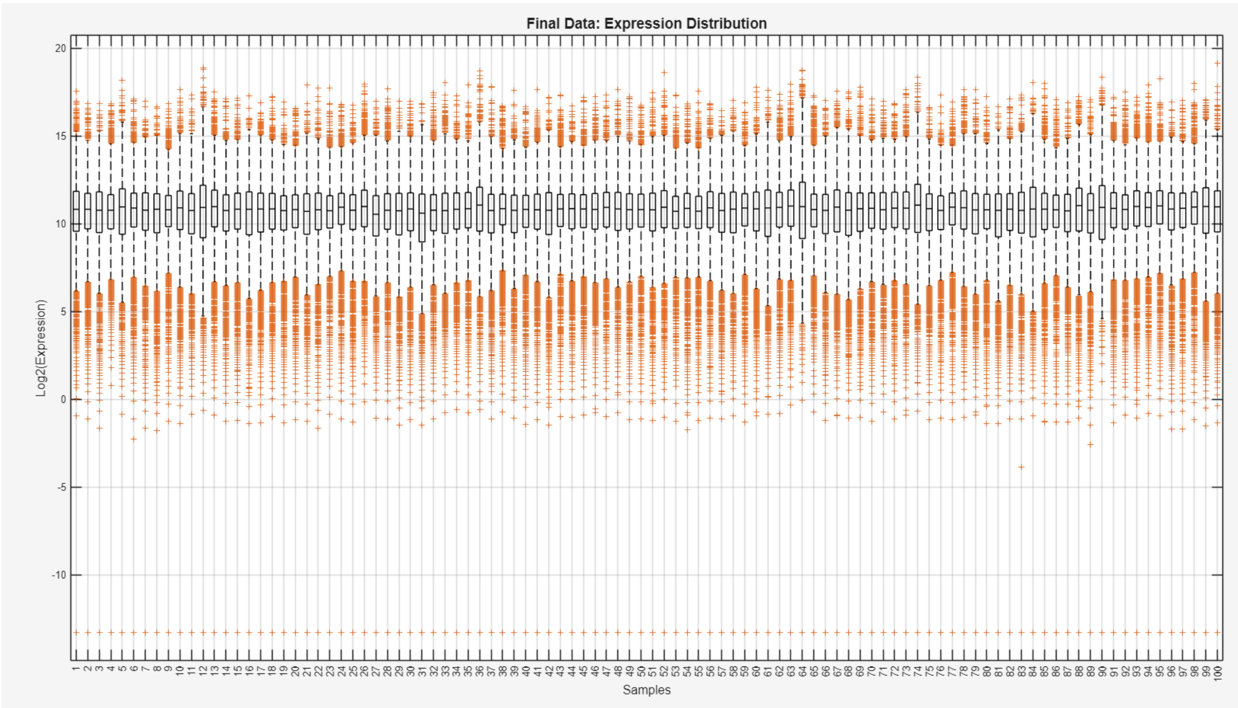


*Figure 11.* Boxplot illustrating expression distribution of samples after filtering out lowly expressed mRNA using a threshold of 0.90

Filtering was conducted to remove lowly expressed genes. A threshold of 0.90 was selected in order to remove mRNA that does not contribute biological significance, rather just noise to the dataset. Moreover, a 0.90 threshold was selected in hopes of not being too strict that genes are removed that are significant biologically.  The small variability seen in Figure 11 is further supported in Figure 12, illustrating the IQR of all 100 samples. As seen in Figure 12, most samples have an IQR ranging between 1.6 and 2.6, illustrating little variability. After removal of lowly expressed genes, Figure 13 illustrates the average Pearson correlation of each sample in relation to all other samples. Only one outlier (TCGA-AX-A0J0)  is seen in Figure 13, however this was not removed considering the good alignment of all samples in relation to one another seen in the boxplot in Figure 11. Moreover, the average Pearson correlation of TCGA-AX-A0J0 is only slightly lower than the alpha threshold. In the final plots, that is Figure 12 and Figure 13, no obvious batch effects were identified such as significant trends between sequential samples or clustering.

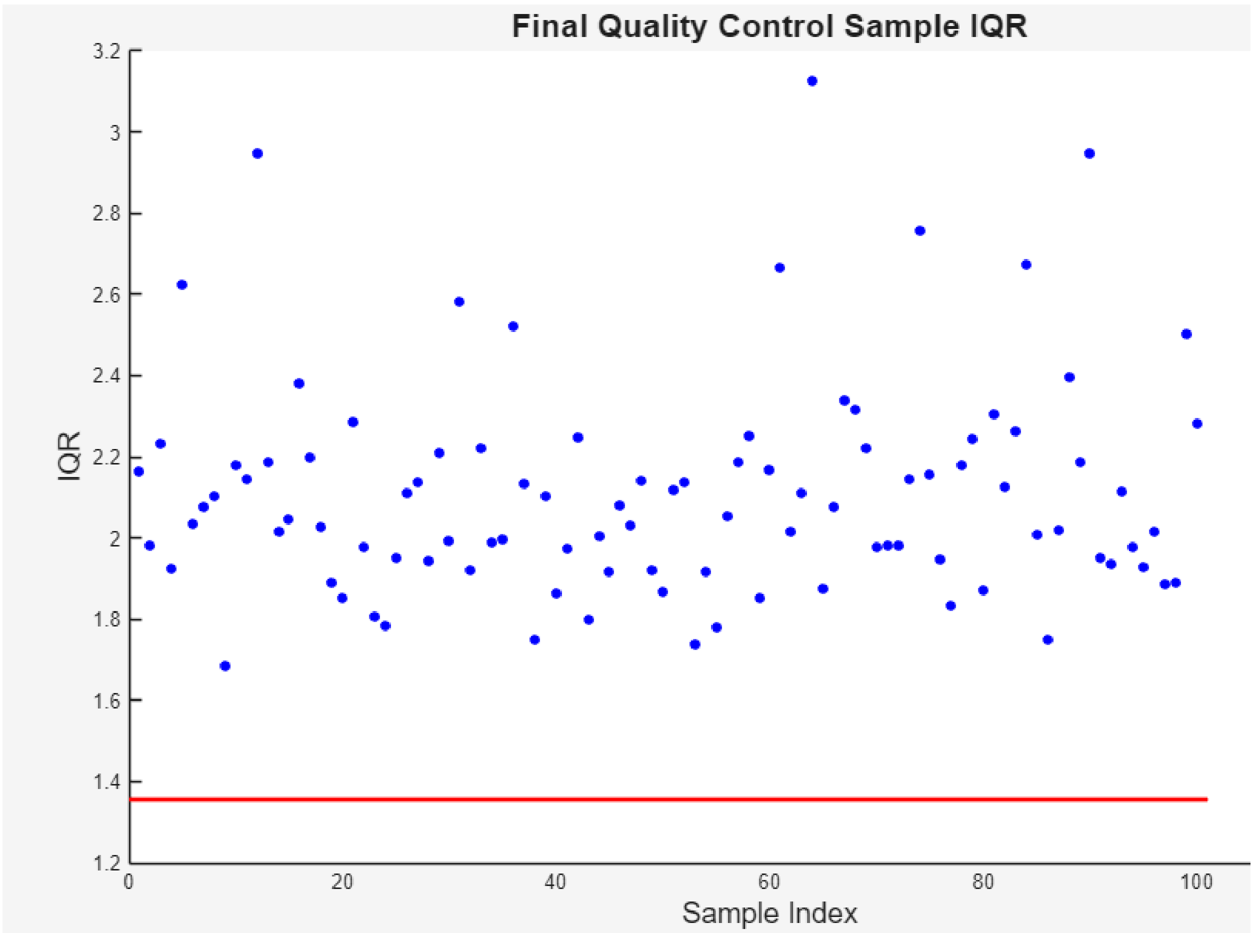


*Figure 12. Scatterplot* illustrating IQR of samples after filtering out lowly expressed mRNA using a threshold of 0.90

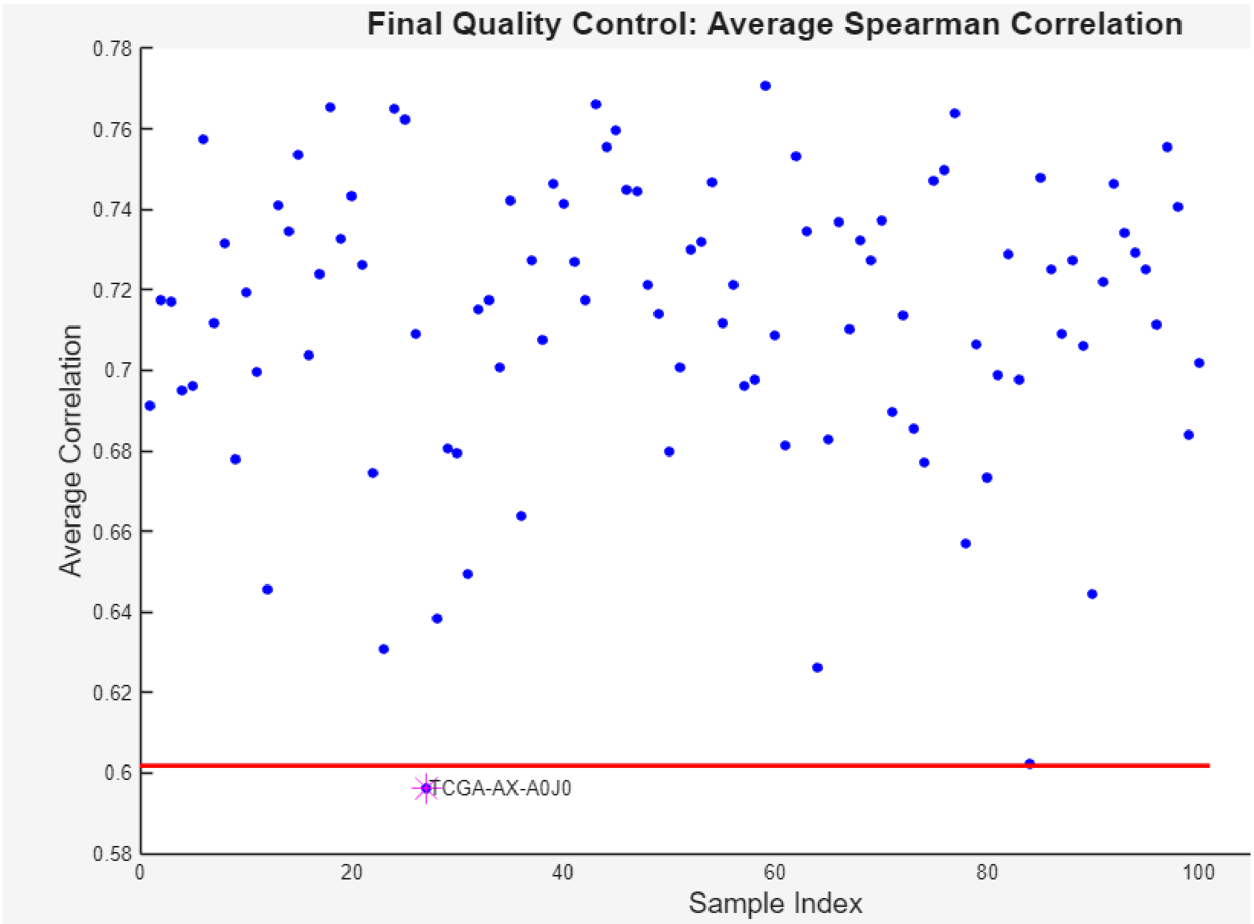


*Figure 13. Scatterplot* illustrating average pearson correlation of samples after filtering out lowly expressed mRNA using a threshold of 0.90

## Feature Selection

The first section here looks at baseline visualziation of t-SNE and different clustering approaches with correlation, Euclidean, Cosine distances and average and complete linkages.

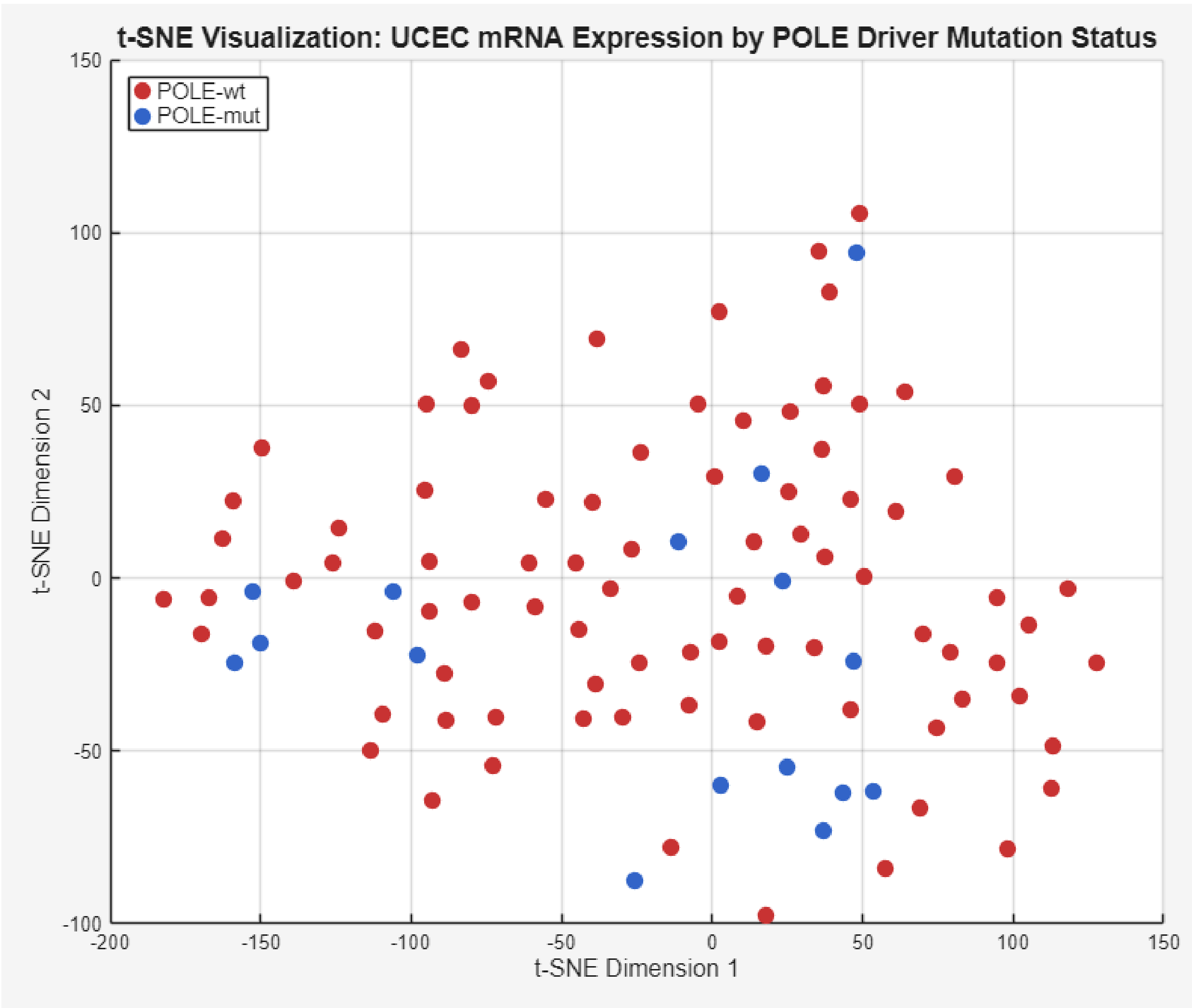


*Figure 14.* t-SNE Visualization for UCEC mRNA Expression

In figure 14, the t-SNE graph has Spearman distance and perplexity set to 13. There are no strong apparent clusters as POLE mutations are relatively rare and there are many other molecular markers relevant for endometrial cancer. Pole mutation and pole wild type are being visualized in lower dimensions. Initial data analysis found 16 mutant types and 84 wildtypes fo the POLE Driver mutation subset.

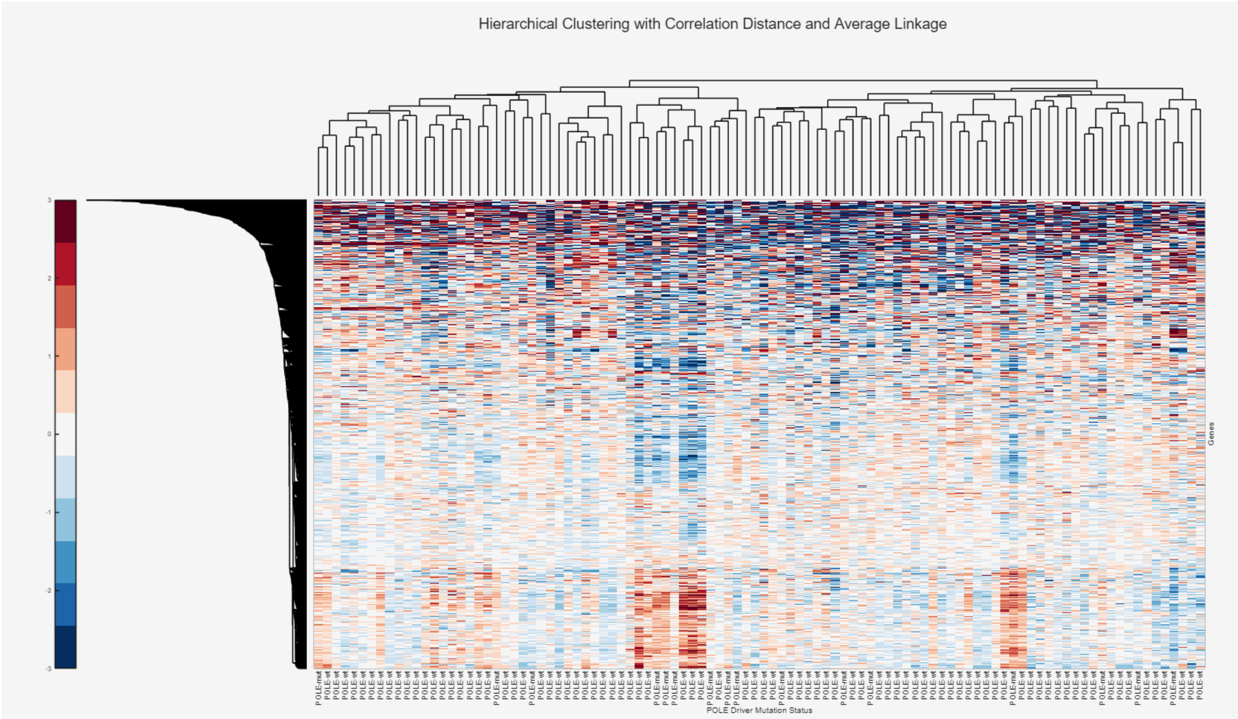


*Figure 15.* Hierarchical Clustering using the correlation distance and average linkage method

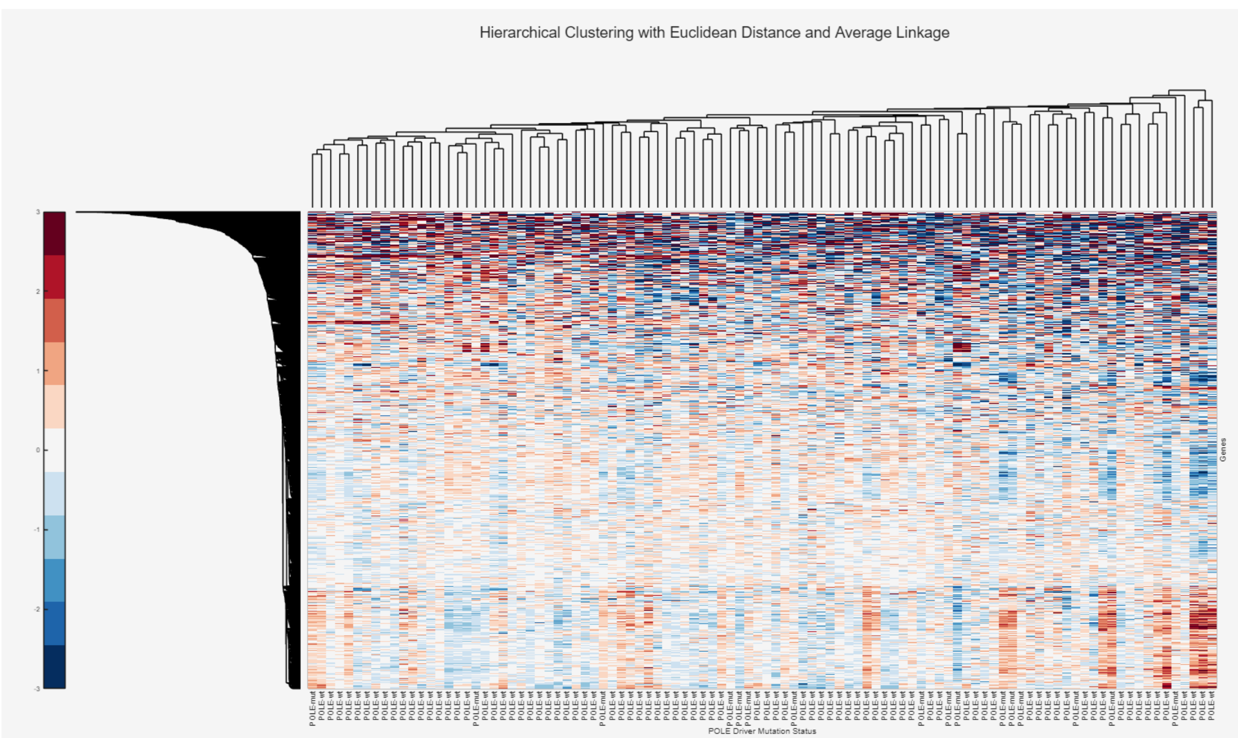


*Figure 16.* Hierarchical Clustering using the euclidean distance and average linkage

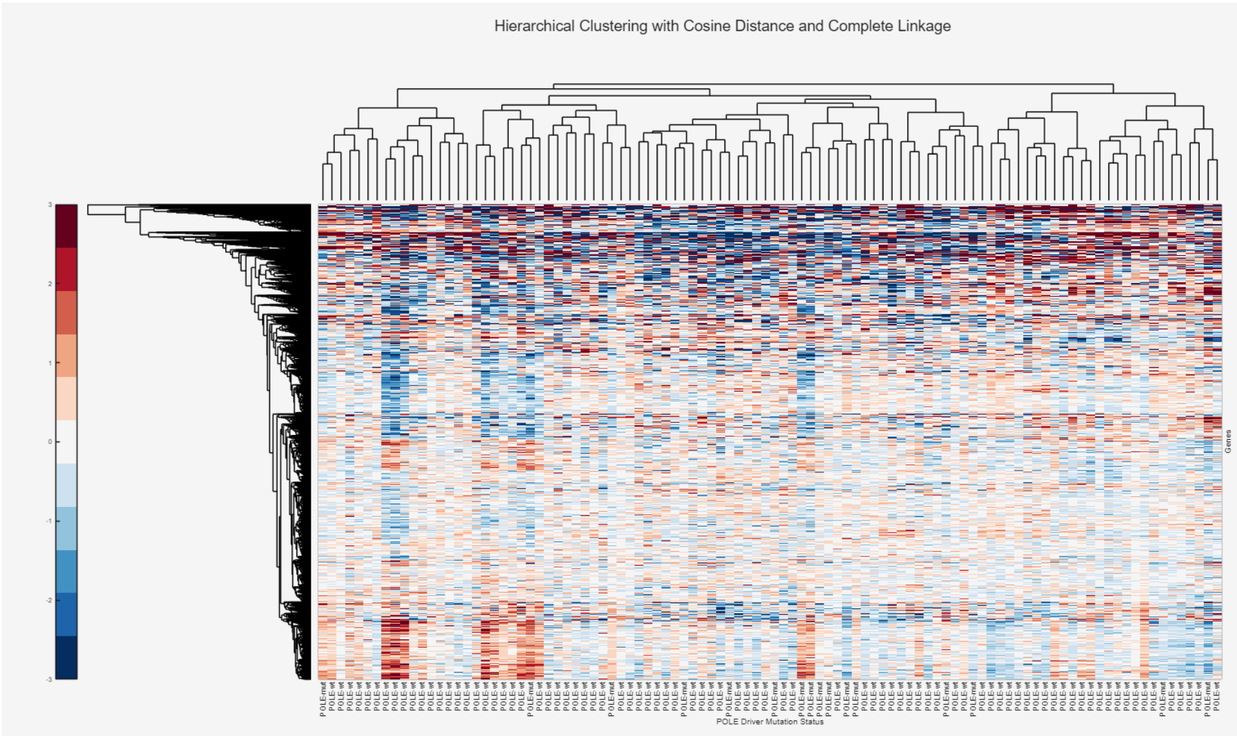


*Figure 17.* Hierarchical Clustering using Cosine distance and complete linkage

There are some clusters that can be spotted with darker bands, as seen in figures 15-17 however, this initial clustering and heatmap visualization is extremely noisy due to the thousands of genes that are being clustered as potential features and the lack of appropriate feature selection. Ultimately, correlation distance was used as it has been proven useful in literature for gene expression analysis by looking at patterns in shape for turned on/off genes and not just magnitude such as Euclidean distance, as well as average linkage for a balanced approach to clusters. They all look quit similar but hopefully correlation may be relevant at picking up clusters for genes relevant to the POLE driver mutation.

**fscchi2 Feature Selection:**

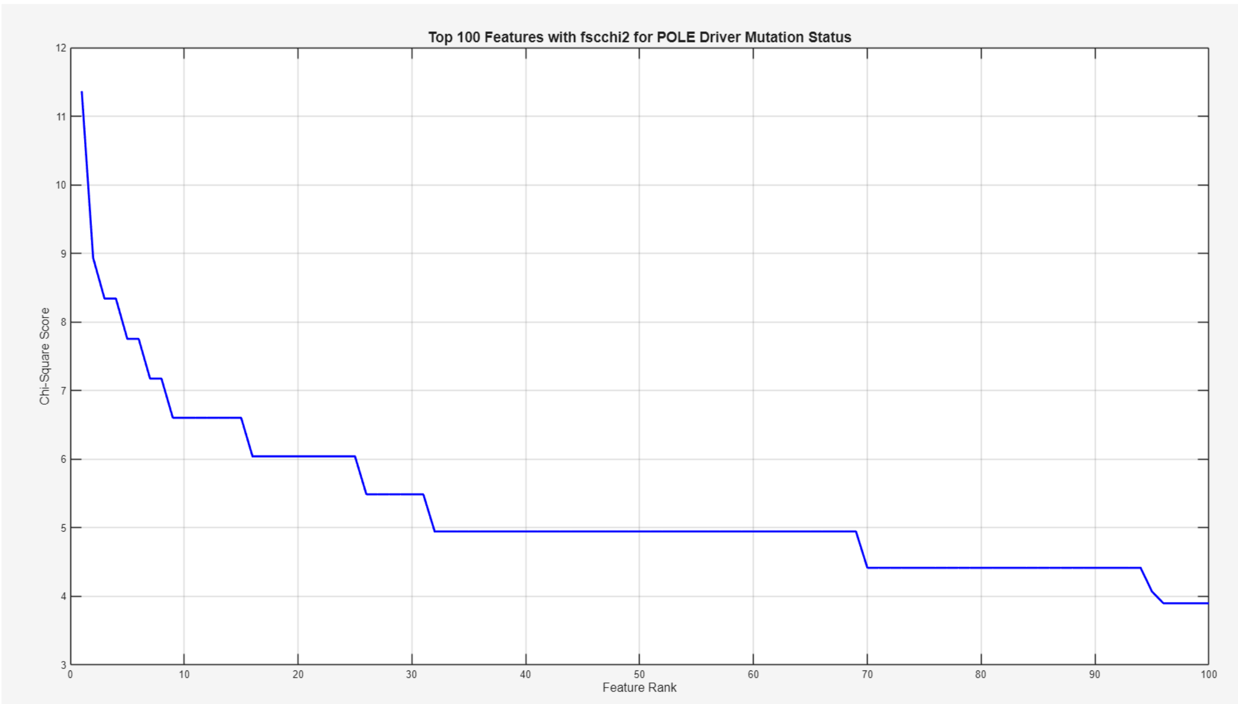


*Figure 18.* Line graph illustrating the top 100 Features Ranked using fscchi2 for POLE Driver Mutation Status

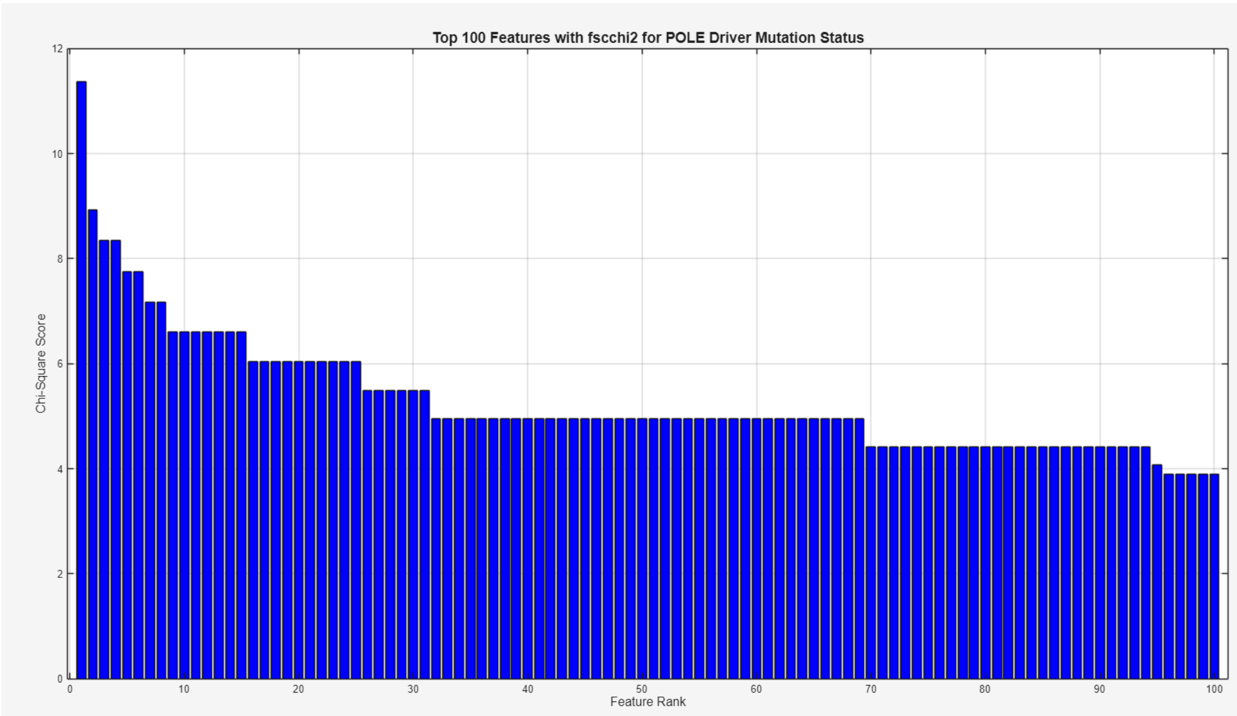


*Figure 19A.* Bar graph illustrating the top 100 features identified using the fscchi feature selection method

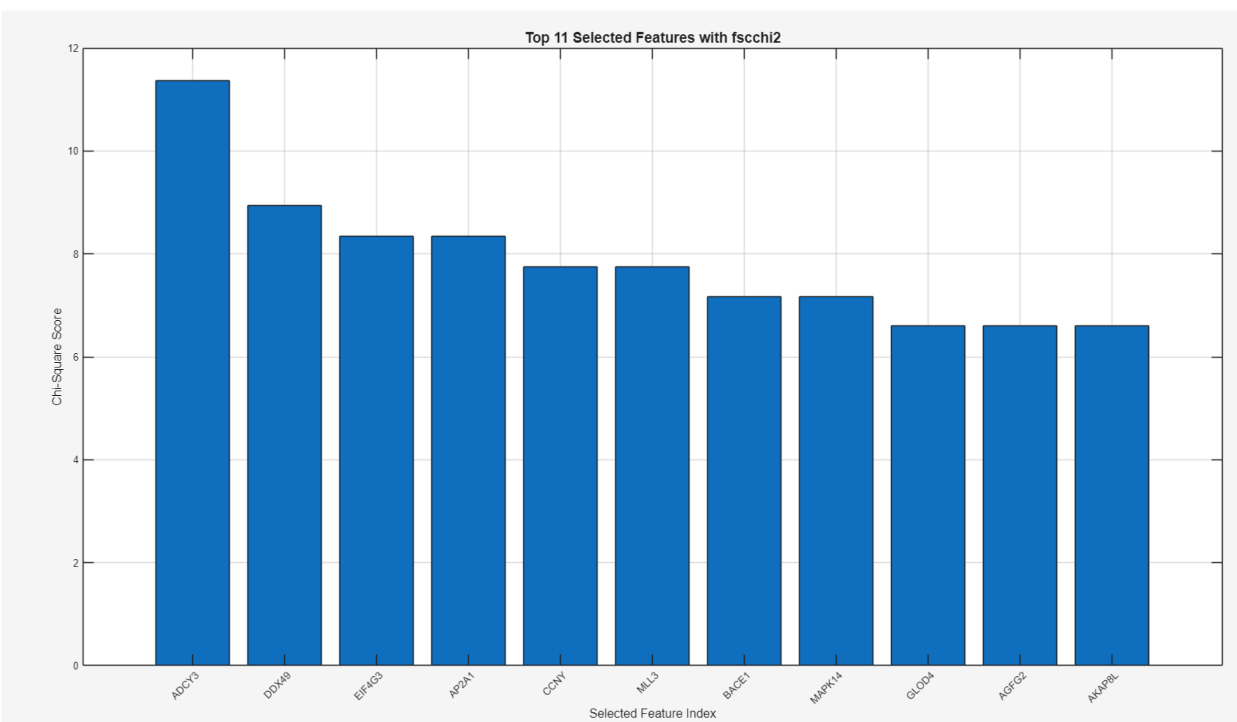


*Figure 19B.* Bar graph of the top 11 selected features using the fscchi feature selection method

As shown in figure 18 and 19A, the top 11 features were selected as suitable elbow for feature selection to balance discrimination vs. overfitting. After the 11th feature elbow, there was a significant and sharp dropoff. Adding additional features would contribute little more than noise for the machine learning portion. Abstaining from adding too many features also helps with overfitting and a worsening clustering structure. The names of the selected mRNA and its corresponding Chi-Square score can be better visualized in FIgure 19B. In all 3 feature selection algorithms experimented with here and below, most elbows plateaurd around 10-20 features and were quite similar in importance for their scores, so 11, 15, and 20 was experimented with as various thresholds to see clustering patterns and balance between discrimination vs. overfitting so poor generalizability does not occur with too many features.

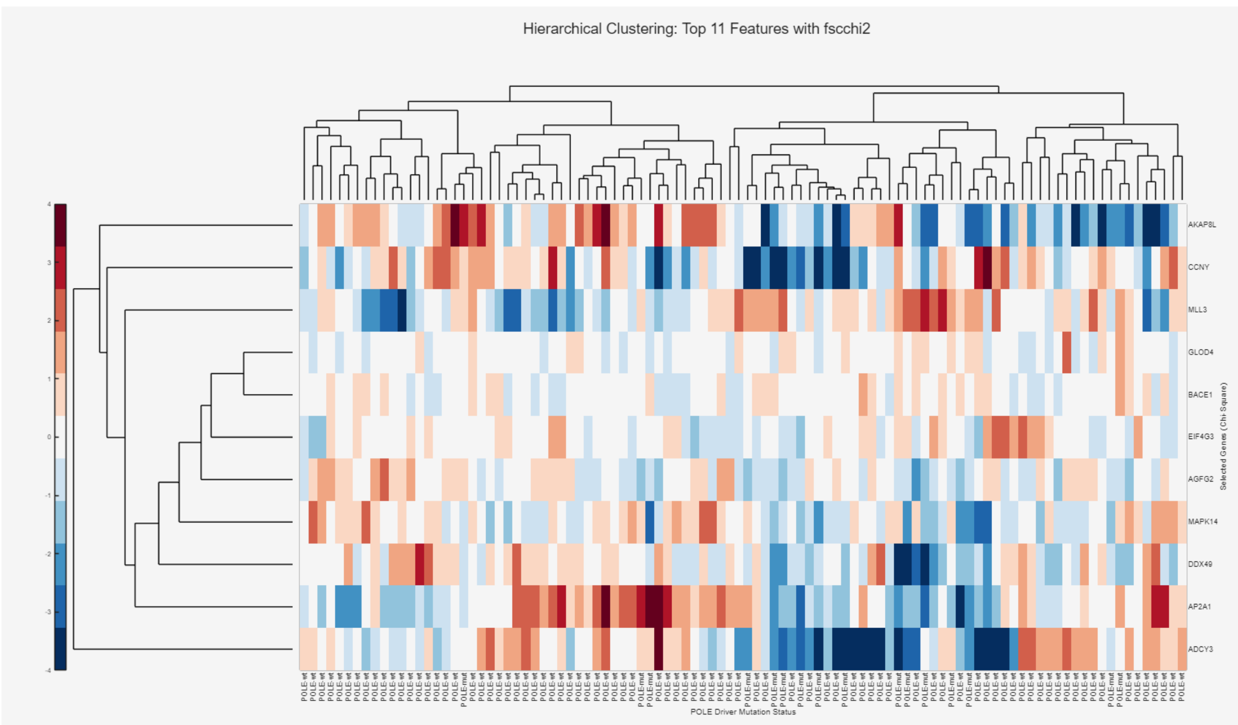


*Figure 20.* Hierarchical clustering after choosing the top 11 features using the fscchi selection method

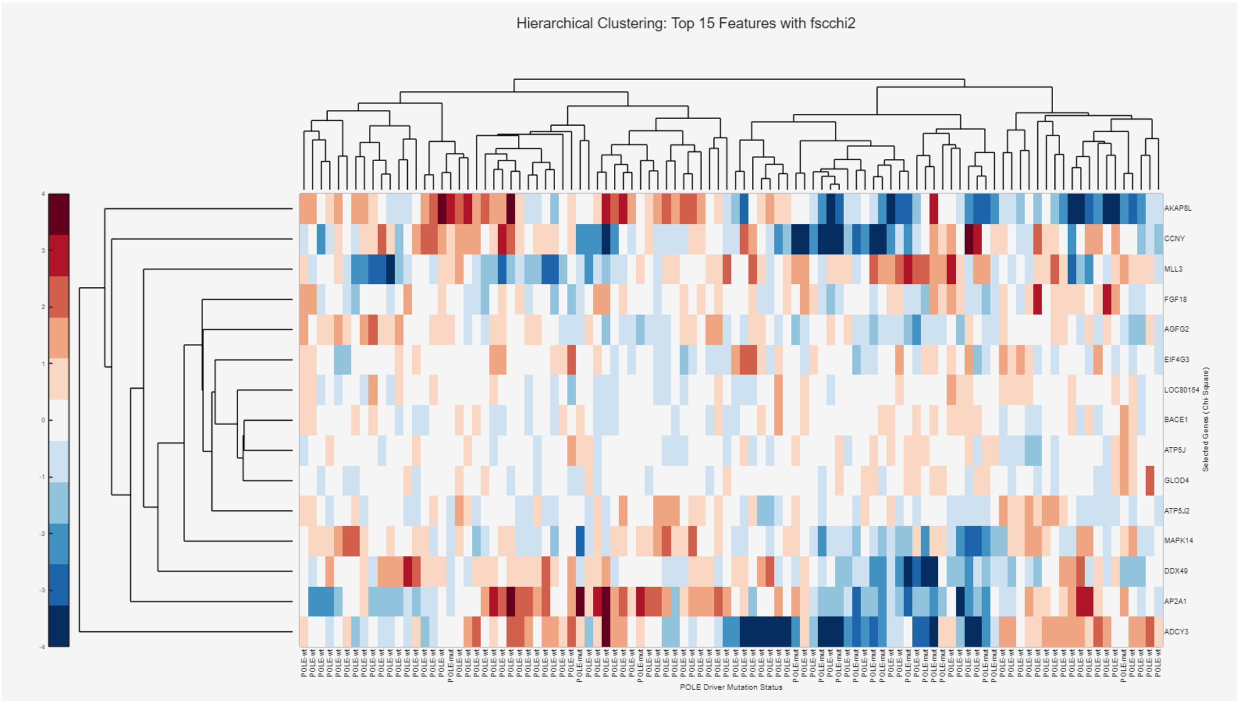


*Figure 21.* Hierarchical clustering after choosing the top 15 features using the fscchi selection method

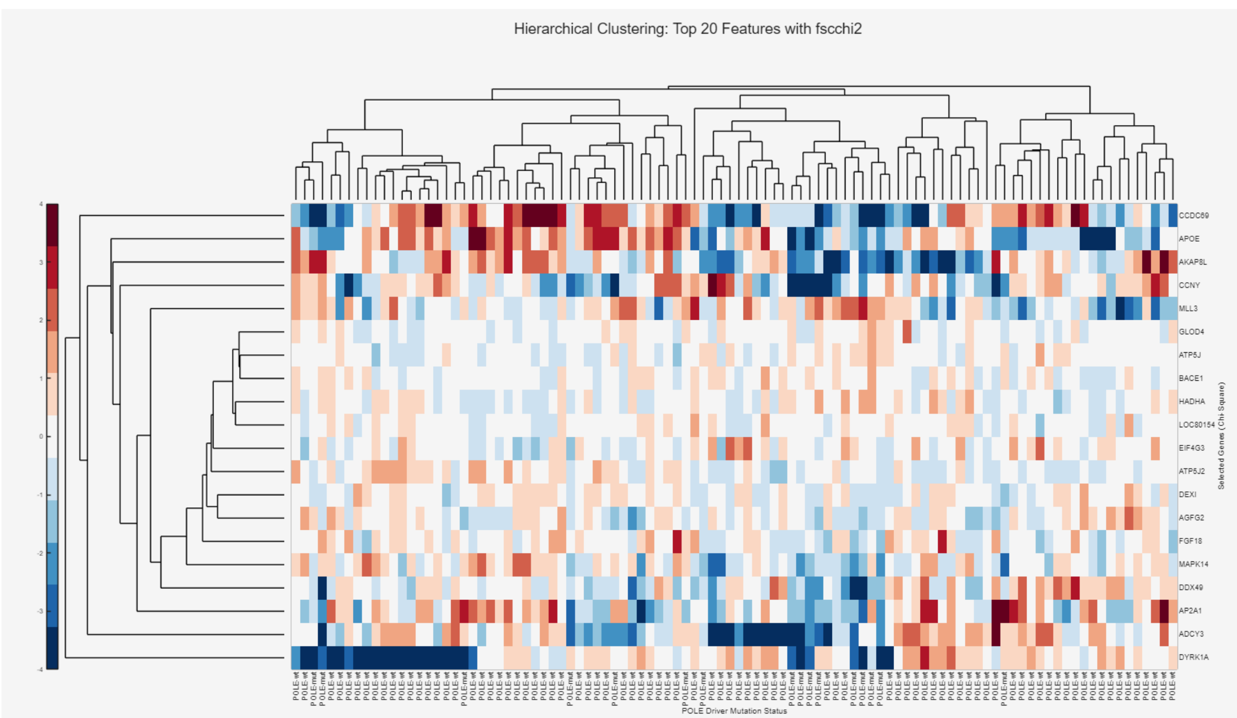


*Figure 22.* Hierarchical clustering after choosing the top 20 features using the fscchi selection method

Figures 20-22 illustrate hierarchical clustering when choosing a different number of features from the fscchi method. Some clusters are seen in each of the three graphs, as demonstrated by darker bands. Hierarchical clustering was also observed using the top 15 and 20 features as there are also dropoffs seen around these features. Nonetheless, incorporating more features on top of the preferred 11 features illustrates greater noise with minimal additional clustering.  When comparing the tSNE plot from before (Figure 14) and after feature selection using fscchi (Figure 23), there is still little clustering seen in the graph when looking at samples with and without the POLE driver mutation.

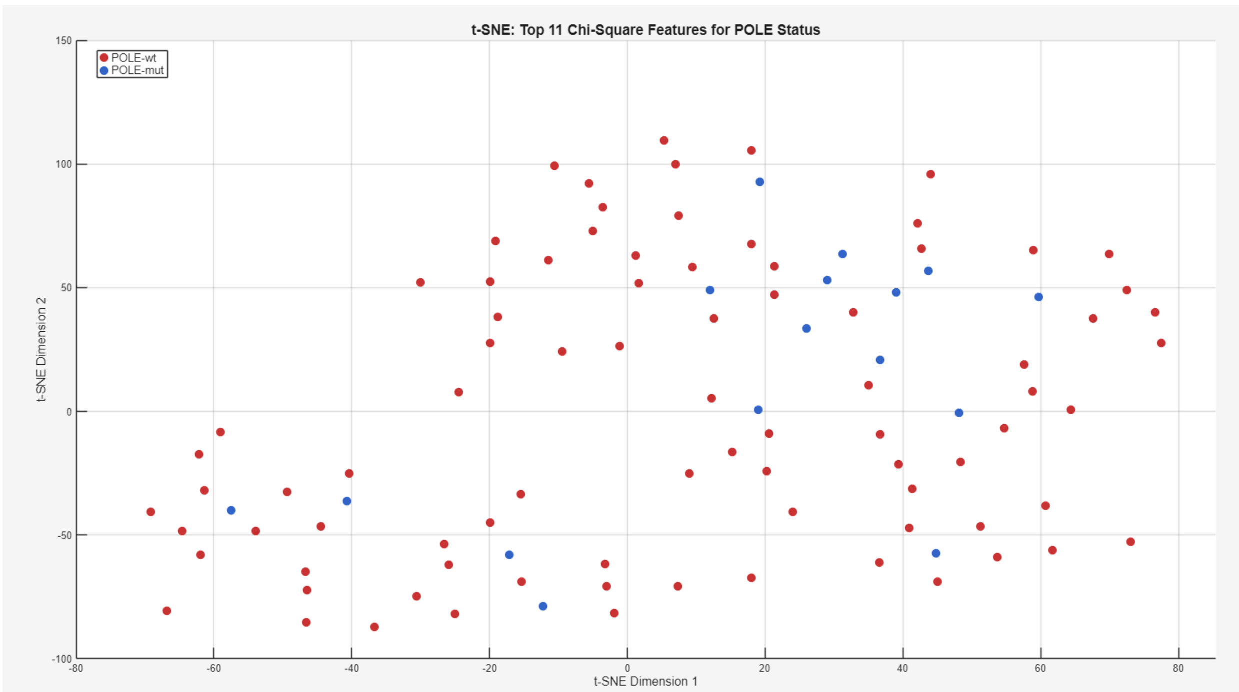


*Figure 23.* tSNE plot after choosing the top 11 features using the fscchi selection method

**fscmrmr Feature Selection:**

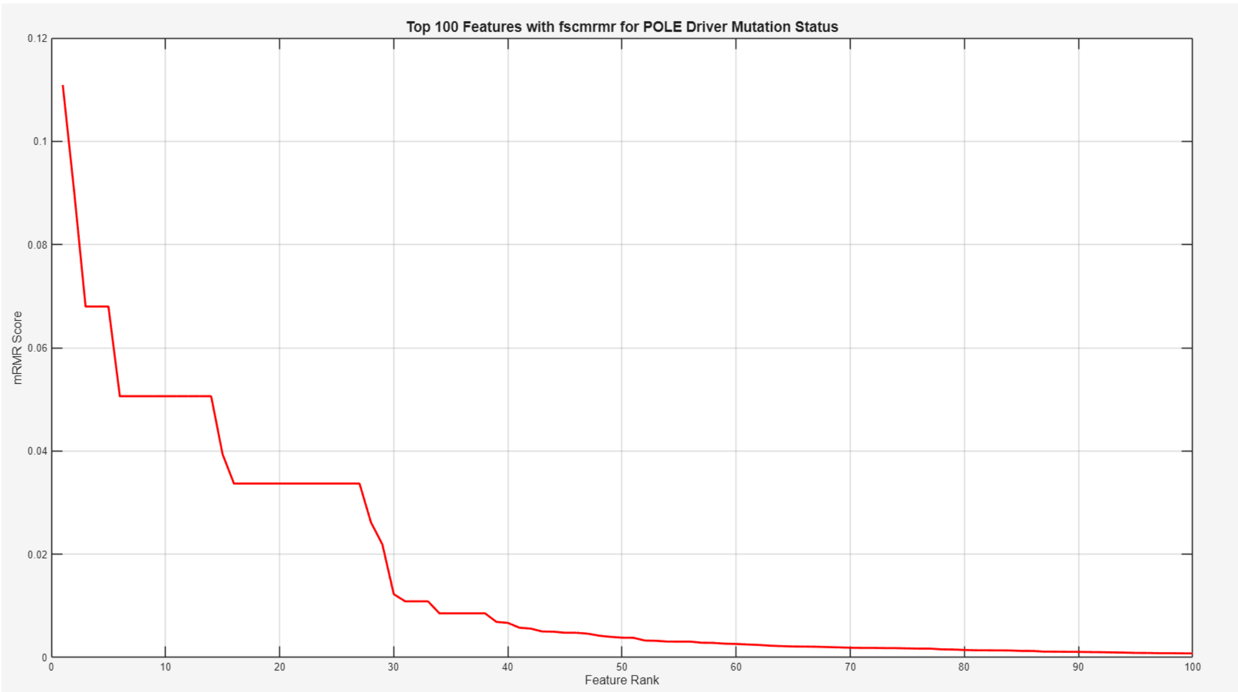


*Figure 24A. Line graph showing* the top 100 Features Ranked using fscmrmr for POLE Driver Mutation Status

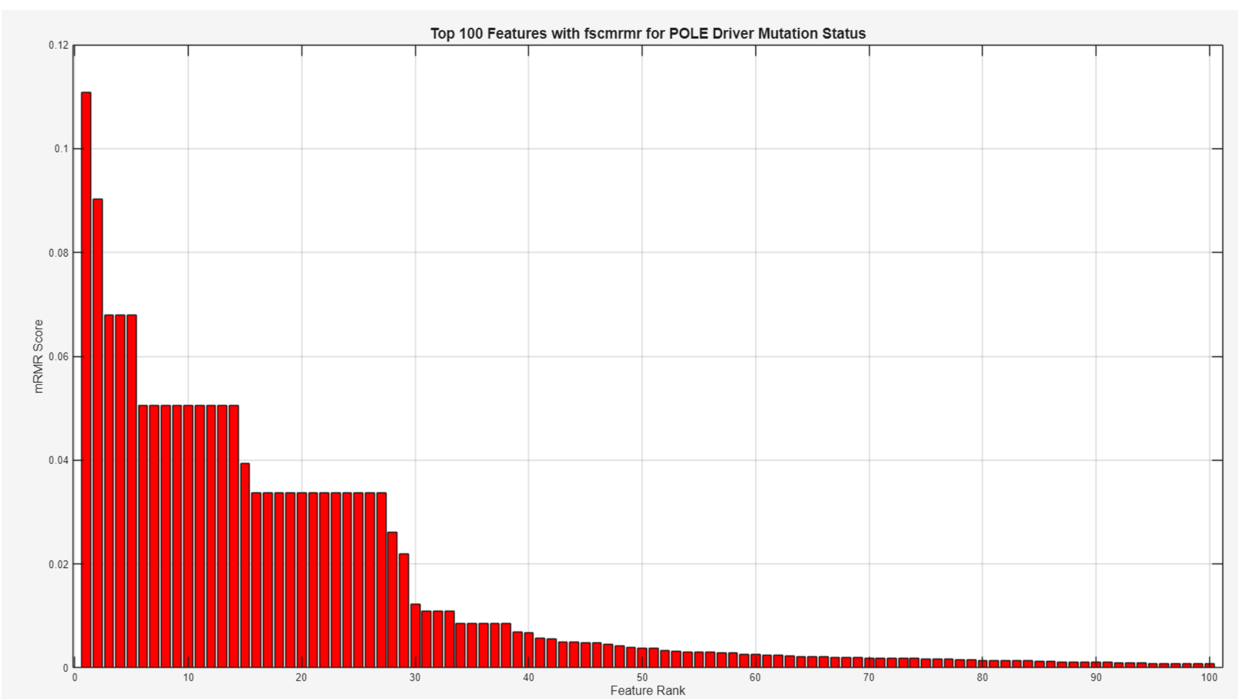

*Figure 24B.* Bar graph showing the top 100 Features Ranked using fscmrmr for POLE Driver Mutation Status

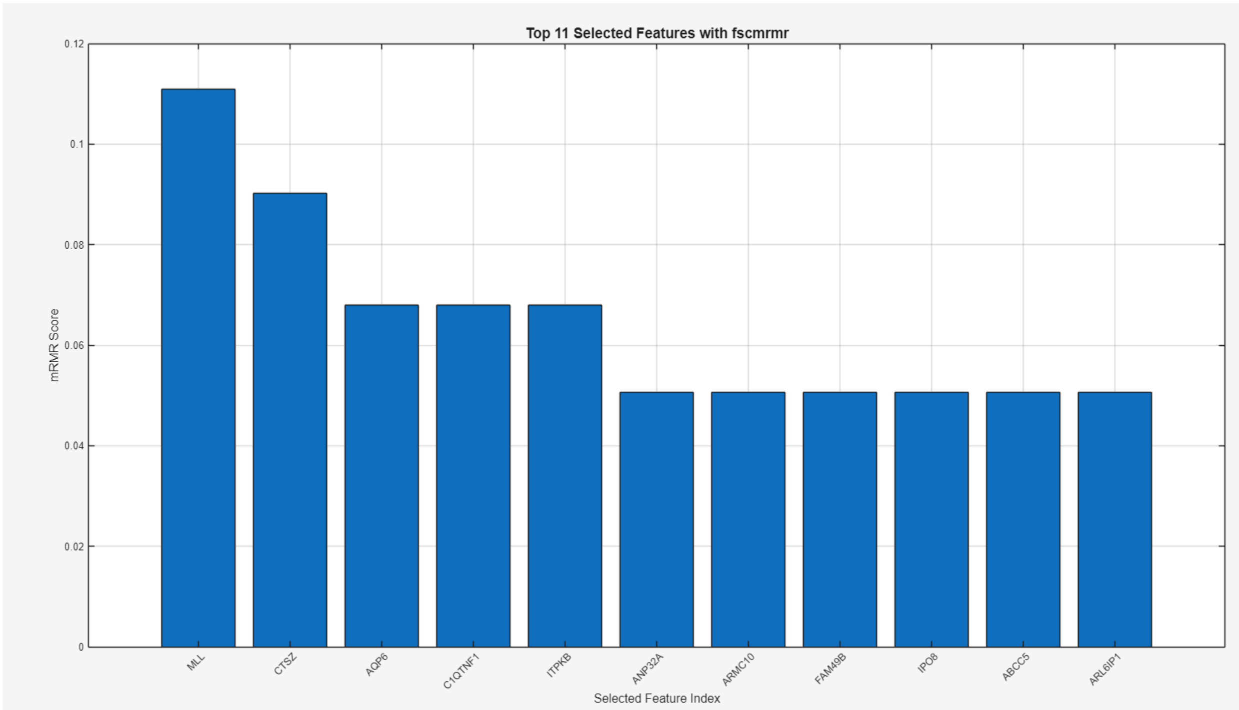


*Figure 25.* Bar Graph of the top 11 selected features using fscmrmr for POLE Driver Mutation Status

Next, feature selection was conducted using the fscmrmr method. In Figure 24A and 24B, there appears to be a drop-off near features 10 to 15, seen in the line graph and further illustrated in the bar graph. After this drop, the mRMR scores appear to flatten and would contribute noise when conducting machine learning. This is also further illustrated below in Figures 26-28 where hierarchical clustering is observed when selecting a different number of features. Figure 26 illustrates some dark banding for genes CTSZ and C1QTQNR1. However, very little additional clustering is seen when increasing the number of features from selecting the top 11, to the top 15 and the top 20. Top 15 and top 20 features were selected to account for different drop-offs seen in the graphs above. As such, this supports the idea that selecting the top 11 features would prevent additional noise or overfitting when conducting machine learning.

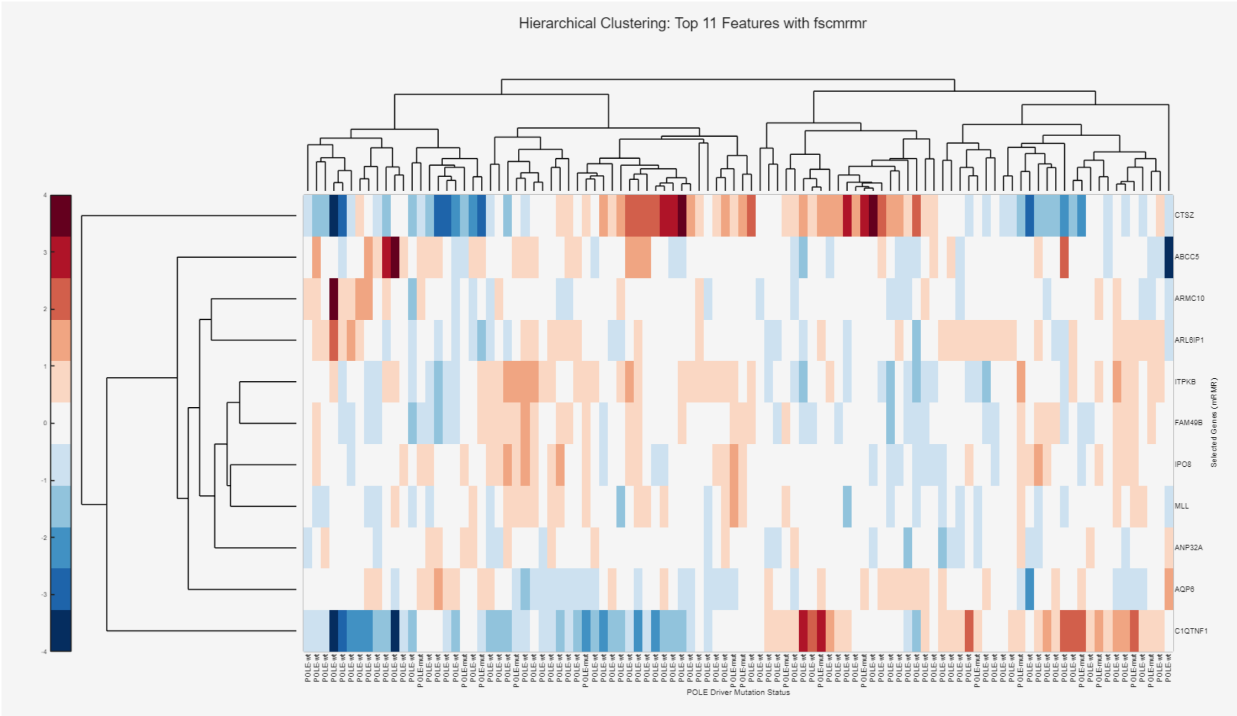


*Figure 26.* Hierarchical clustering after choosing the top 11 features using the fscmrmr selection method

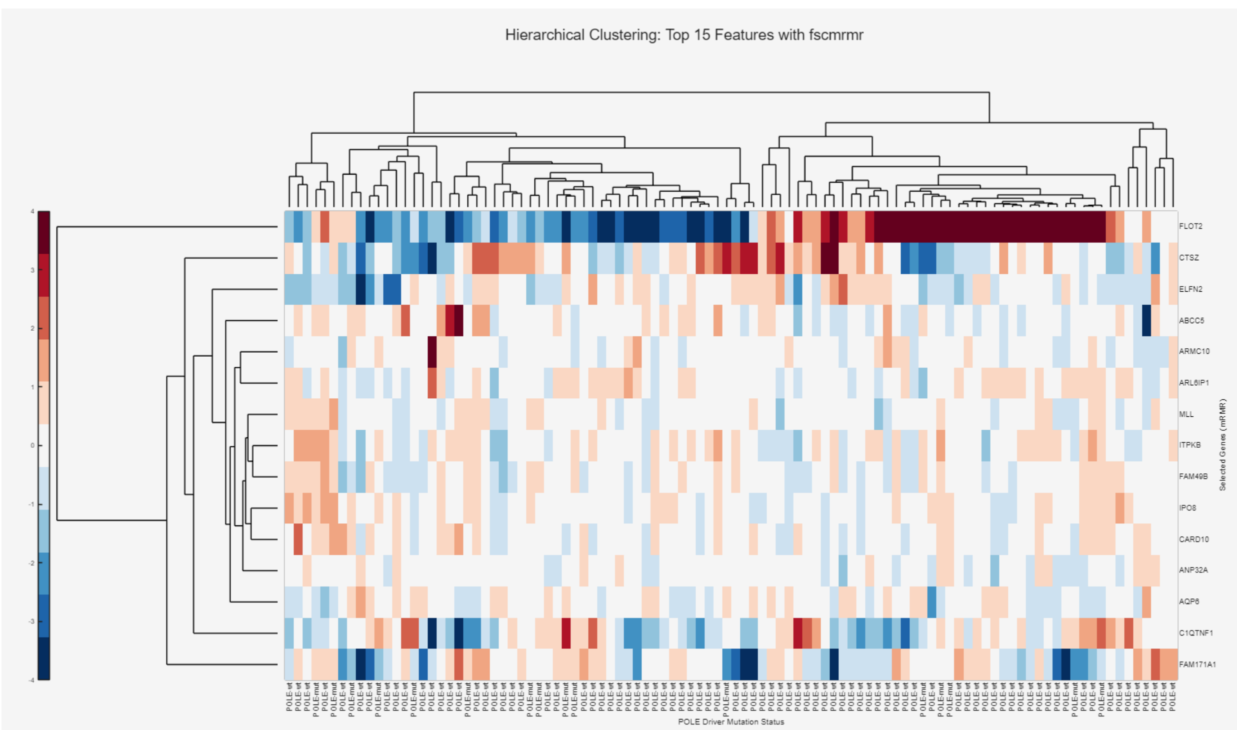


*Figure 27.* Hierarchical clustering after choosing the top 15 features using the fscmrmr selection method

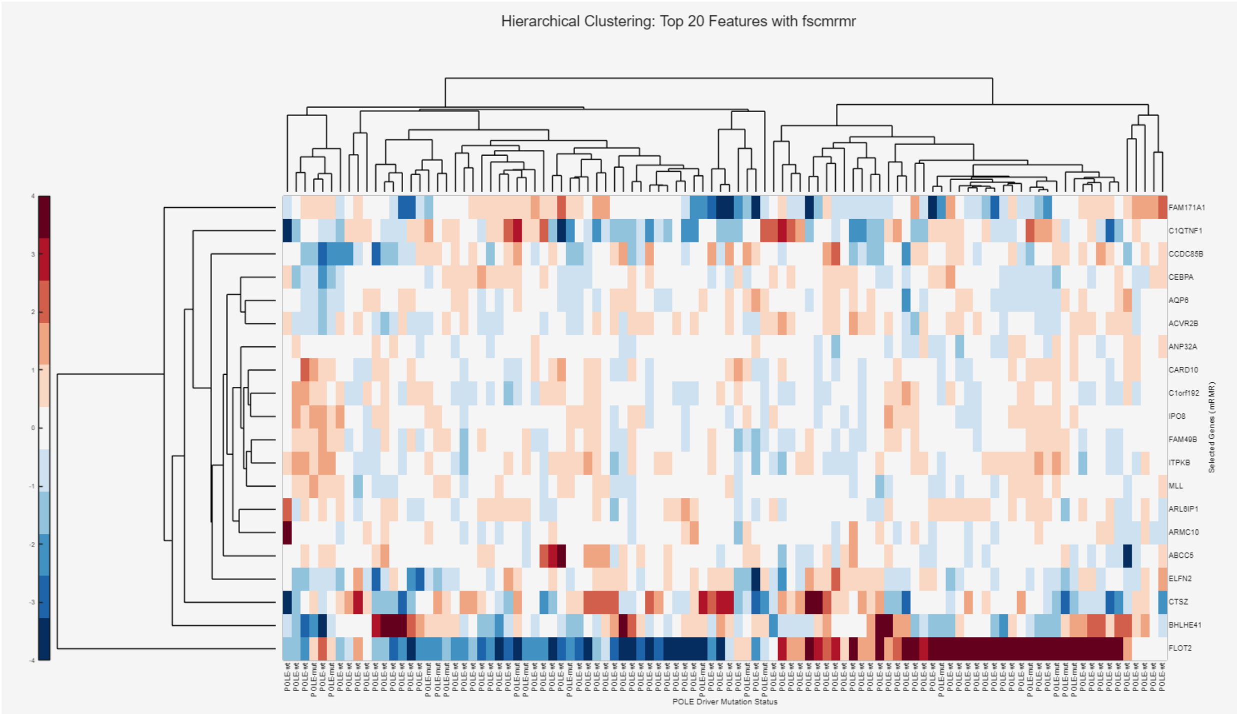


*Figure 28.* Hierarchical clustering after choosing the top 20 features using the fscmrmr selection method

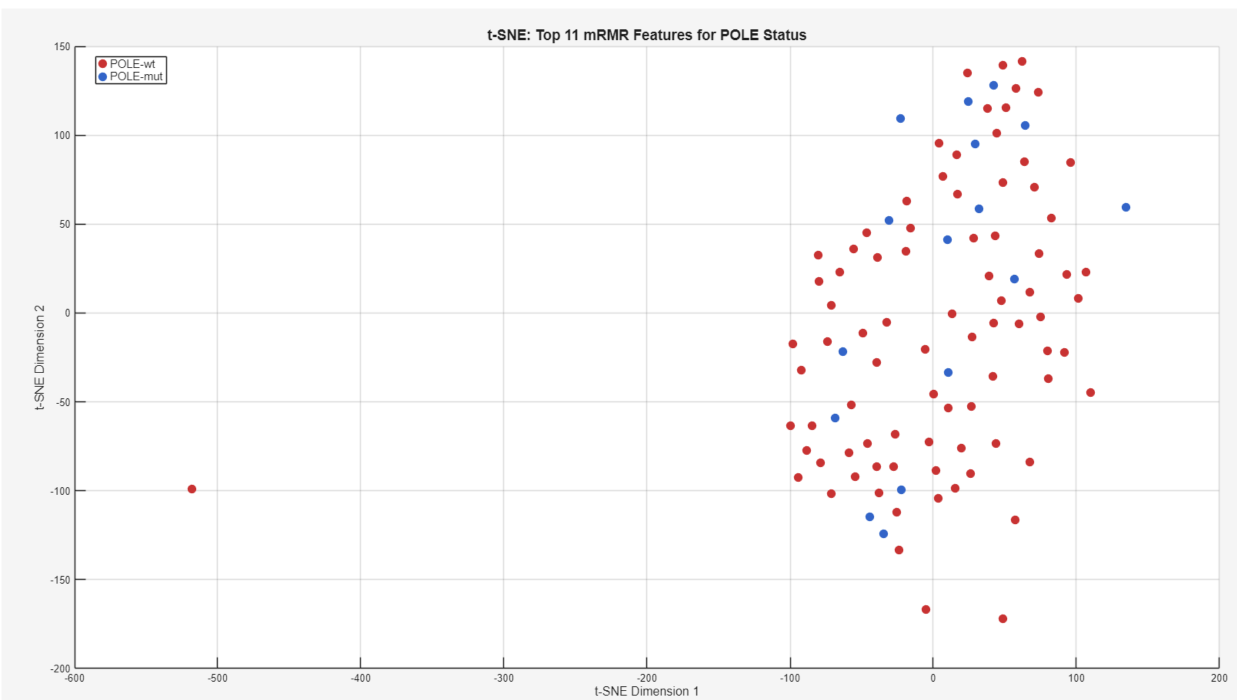


*Figure 29* tSNE plot after choosing the top 11 features using the fscmrmr selection method

Similar to before feature selection using fscmrmr was conducted, very little clustering was revealed in the tSNE plot between samples with and without the mutation after choosing the top 11 features.

**relieff Feature Selection**

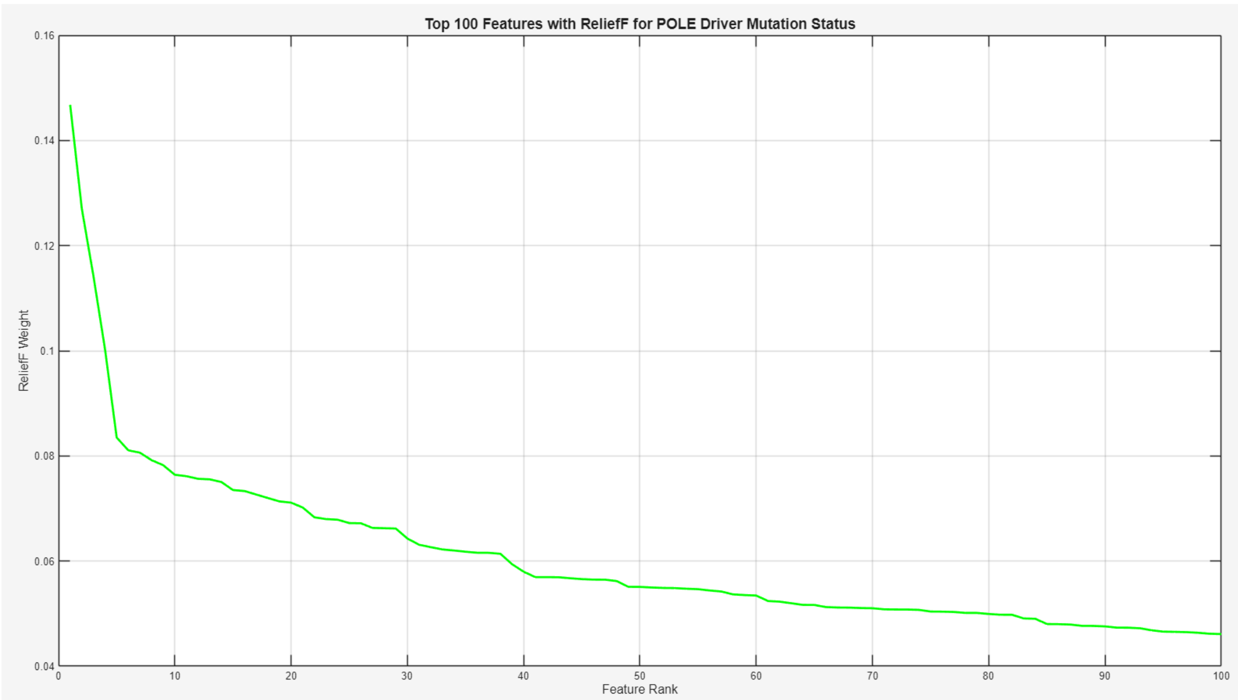


*Figure 30A.*Line graph showing top 100 Features Ranked using ReliefF for POLE Driver Mutation Status

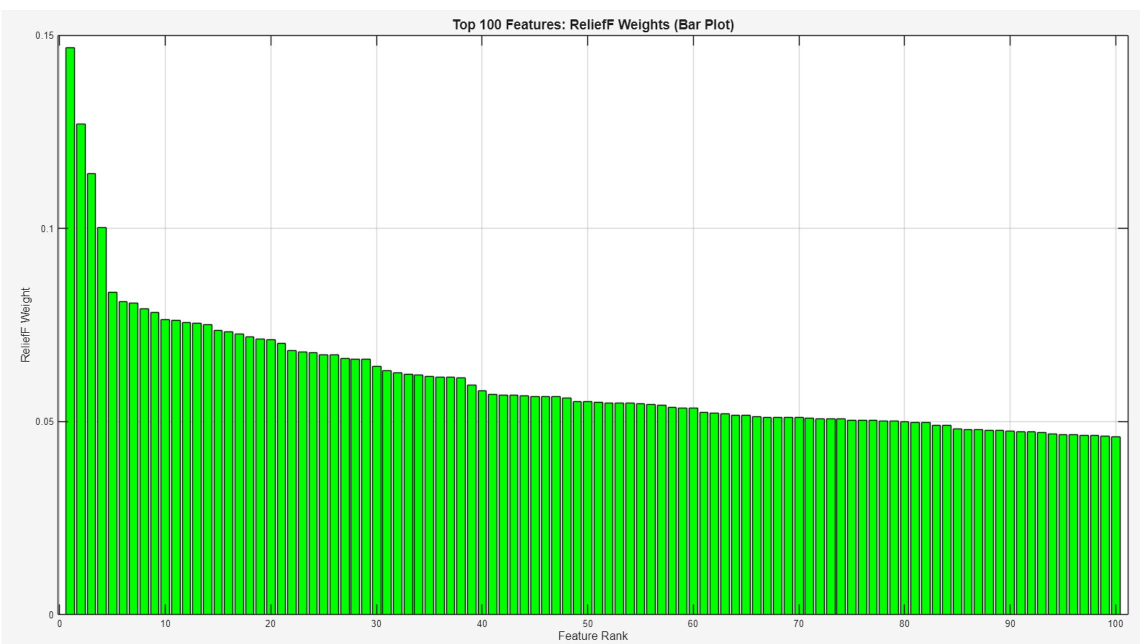


*Figure 30B.* Bar graph showing top 100 Features Ranked using ReliefF for POLE Driver Mutation Status

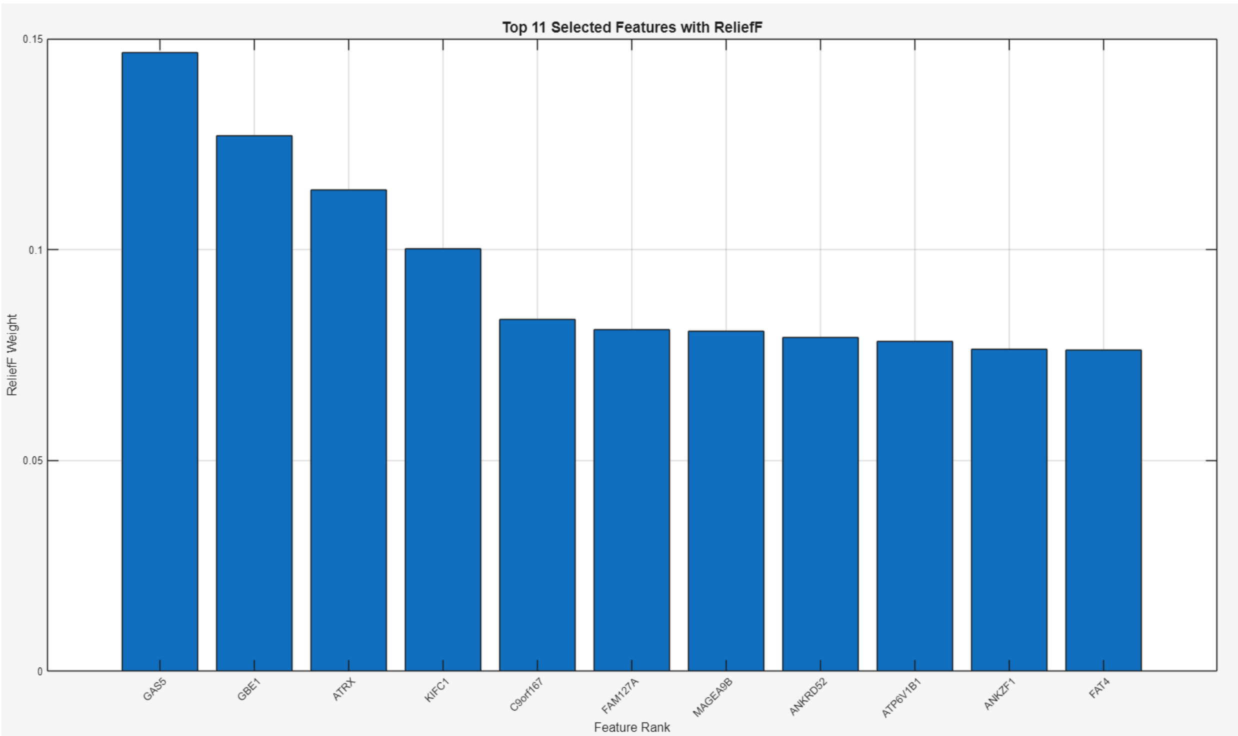


*Figure 31.* Bar Graph of the top 11 selected features using ReliefF for POLE Driver Mutation Status

When looking at feature selection using ReliefF, both Figure 30A and 30B illustrate a dropoff near feature 5 and a steady decrease afterwards. To avoid overfitting and ensure strong discrimination, more features were decided to be used when analyzing features using hierarchical clustering and tSNE plots. When selecting the top 11 features, hierarchical clustering seen in Figure 32 illustrated clear banding in comparison to other feature selection methods. Genes such as ATRX, ATP6V1B1, GBE1, and GAS5 demonstrate clear banding. Higher expression of these genes were associated with mutation (illustrated in dark red banding) and lower expression of these genes were associated with no mutation (illustrated in dark blue banding).

To prevent overfitting and to balance discrimination, we also tested hierarchical clustering with the top 15 and top 20 features revealed by ReliefF, as seen in Figures 32 and 33. The addition of more features contributed more noise, with less overall banding. Moreover, there was less clear discriminatory power illustrated by the additional genes as there was no clear high or low expression seen in individuals with or without the mutation.

Moreover, the tSNE clustering using the top 11 features revealed by ReliefF illustrated between clustering overall, particularly of the samples containing the ROLE driver mutation, in comparison to the other feature selection methods where mutation samples were more scattered. Nonetheless, the tSNE plots are still relatively noisy, which may be due to issues with using mRNA expression to identify POLE driver mutations.

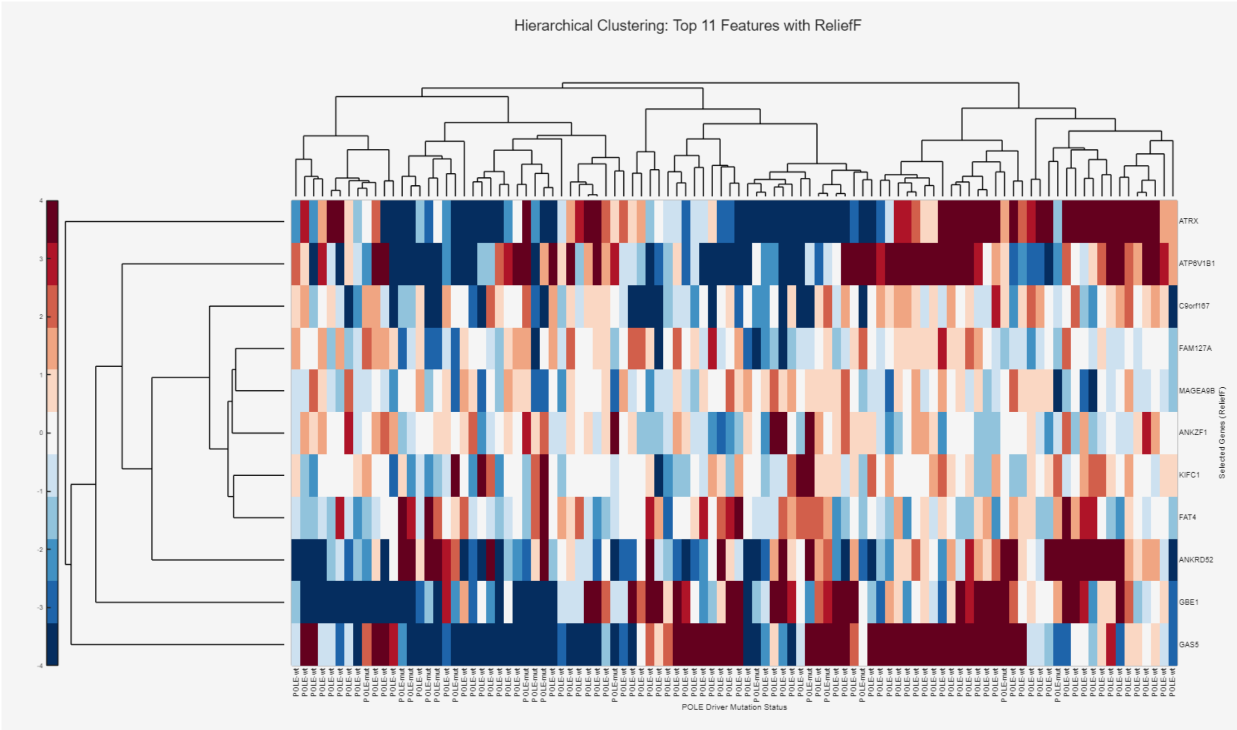


*Figure 32.* Hierarchical clustering after choosing the top 11 features using the ReliefF selection method

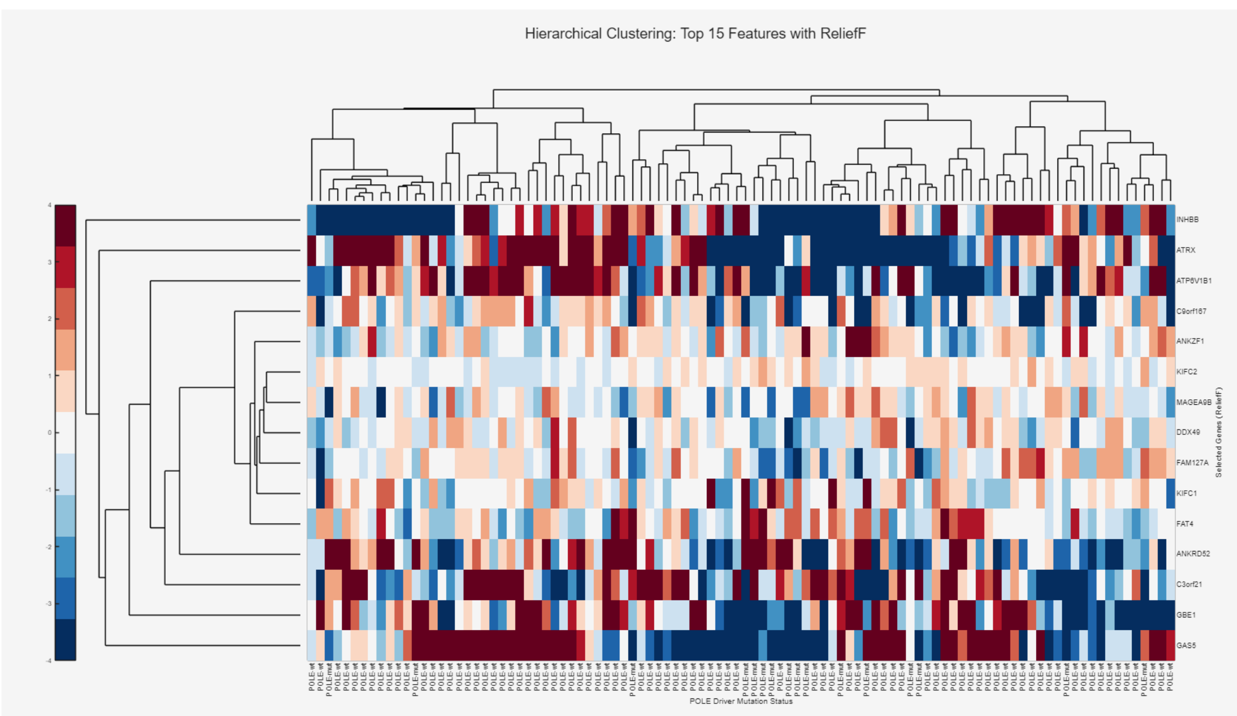


*Figure 33.*Hierarchical clustering after choosing the top 15 features using the ReliefF selection method

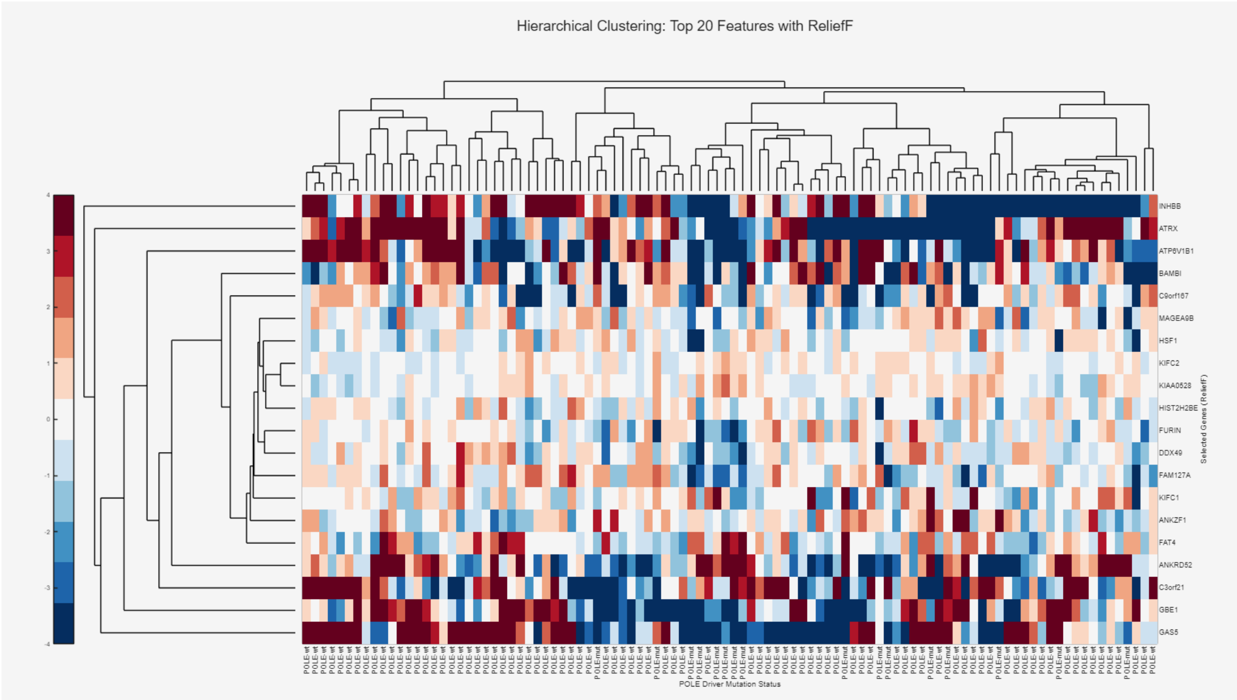


*Figure 34.*Hierarchical clustering after choosing the top 20 features using the ReliefF selection method

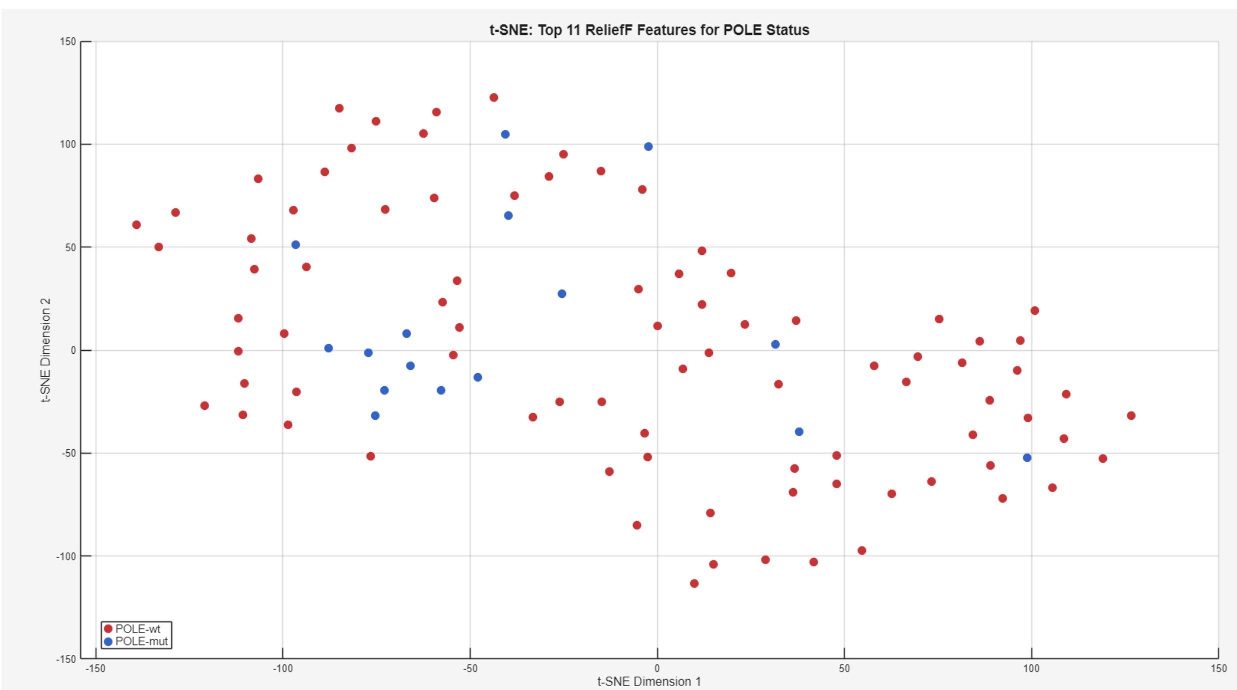


*Figure 35.* tSNE plot after choosing the top 11 features using the ReliefF selection method

Considering the better clustering seen in both hierarchical clustering and the tSNE plot, as well as a reduction of noise, the top 11 features using the ReliefF feature selection method was chosen and utilized in the machine learning portion to create a classifier. Ultimately, all t-SNE plots after feature selection were quite noisy, inherent of the challenges of using solely mRNA expression for mutation patterns, but the t-SNE here looked cleaner with clearer separation and balanced compared to noisier graphs above.

**<u>Classification Analysis</u>**

We now move on to our machine learning portion. 5-fold cross-validation was used to give a solid estimate of model performance regarding overfitting, where the dataset is divided into k equal parts and trained on k-1 folds and validated on the remaining one, repeating the process. k= 5 is a solid balance between performance and variance reduction. An 80/20 split was used with 80% on the training data and 20% for the validation hold-out so the models have enough samples to learn patterns while also maintaining decent generalizability.

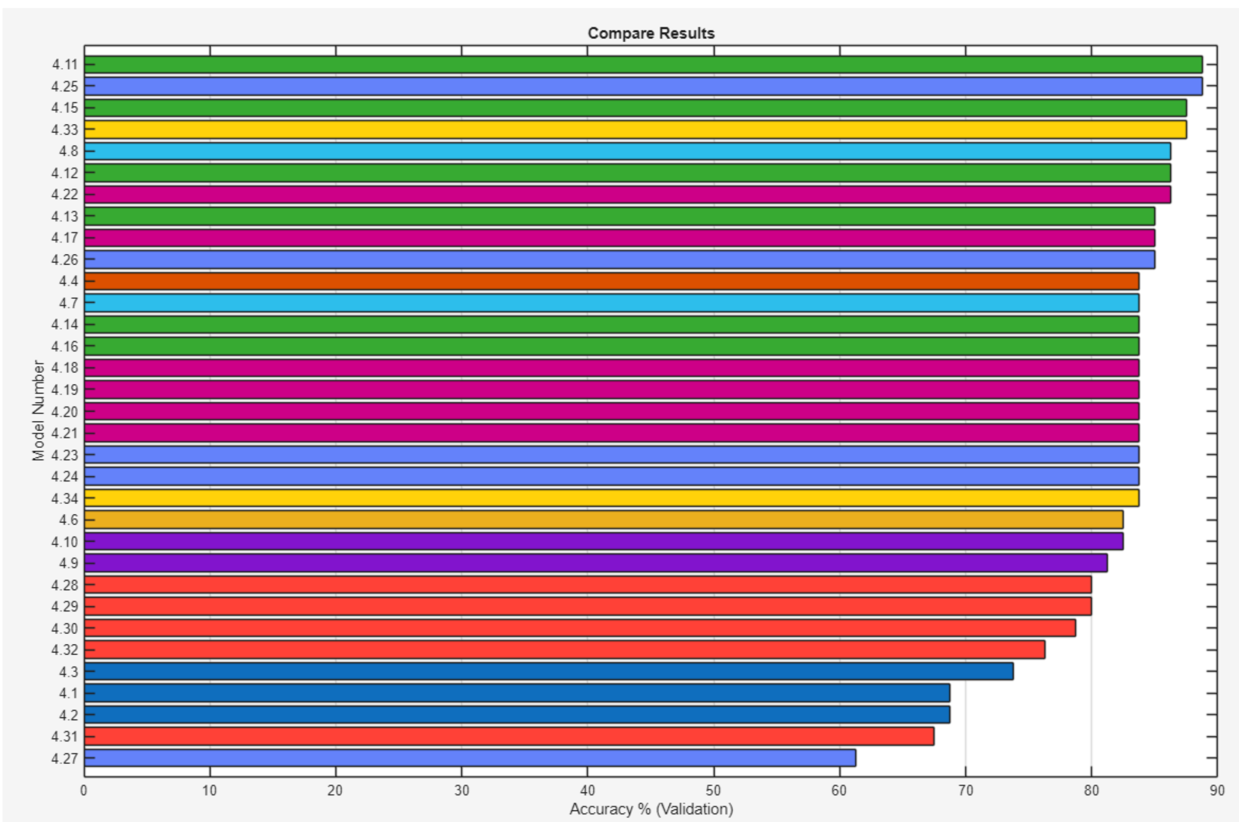


*Figure 36.* Ranking of validation accuracy of all models, with SVMs in green ranking as the strongest discriminating models, followed by ensembles with subspace discriminant in blue. Neural networks and trees (in red and dark blue) tended to perform quite poorly. KNN models (in purple) ranked in the middle in terms of discrimination. The strongest performing model was the linear SVM with a validation

accuracy of 88.8% along with an ensemble with subspace discriminant with an accuracy of 88.8%. The poorest performing model was an ensemble with RUSBoosted Trees with a validation accuracy of 61.3%.

**Table 1.** Ranking of machine learning models for classification of POLE driver mutation predictions based on top 11 selected mRNA features.

| Model | Type | Validation Accuracy |
|---|---|---|
| 4.11 | SVM | 88.75% |
| 4.25 | Ensemble | 88.75% |
| 4.33 | Kernel | 87.50% |
| 4.15 | SVM | 87.50% |
| 4.8 | Efficient Linear SVM | 86.25% |
| 4.12 | SVM | 86.25% |
| 4.22 | KNN | 86.25% |
| 4.13 | SVM | 85.00% |
| 4.17 | KNN | 85.00% |
| 4.26 | Ensemble | 85.00% |
| 4.4 | Discriminant | 83.75% |
| 4.7 | Efficient Logistic Regression | 83.75% |
| 4.14 | SVM | 83.75% |
| 4.16 | SVM | 83.75% |
| 4.18 | KNN | 83.75% |
| 4.19 | KNN | 83.75% |
| 4.2 | KNN | 83.75% |
| 4.21 | KNN | 83.75% |
| 4.23 | Ensemble | 83.75% |
| 4.24 | Ensemble | 83.75% |
| 4.34 | Kernel | 83.75% |
| 4.6 | Binary GLM Logistic Regression | 82.50% |
| 4.1 | Naive Bayes | 82.50% |
| 4.9 | Naive Bayes | 81.25% |

| | | |
|---|---|---|
| 4.28 | Neural Network | 80.00% |
| 4.29 | Neural Network | 80.00% |
| 4.3 | Neural Network | 78.75% |
| 4.32 | Neural Network | 76.25% |
| 4.3 | Tree | 73.75% |
| 4.1 | Tree | 68.75% |
| 4.2 | Tree | 68.75% |
| 4.31 | Neural Network | 67.50% |
| 4.27 | Ensemble | 61.25% |

Above, classes of support vector machines, ensembles, and K-nearest neighbours tended to perform the best while neural networks and decision trees performed poorly. Ultimately, a large issue with this classification task is the severe class imbalance present in the dataset, with 84 POLE-wildtype samples versus only 16 POLE-mutant samples (16:84 ratio). This imbalance limits classification performance, as models can achieve 84% accuracy simply by predicting the majority class (wildtype) for all samples. The observed PR-AUC values reflect this challenge: precision-recall curves for the "Yes" (POLE-mutant) class ranged from 0.55-0.60 in some models, indicating poor ability to correctly identify true positive POLE mutations without generating substantial false positives, while the "No" (wildtype) class achieved PR-AUC values above 0.89, which is expected given its overwhelming prevalence in the dataset. ROC-AUC values hovered around 0.70 for both classes in some models, suggesting only modest discrimination ability above random chance where AUC is 0.5. While ROC curves are often reported for binary classification, they can be misleading in highly imbalanced scenarios because they treat false positives and false negatives symmetrically. In our case, precision-recall metrics provide a more honest assessment of model performance for the minority POLE-mutant class.

Moreover, we believe one of the limitations of this data analysis is that mRNA expression has limited discriminative ability for POLE driver mutation status. POLE mutations affect DNA polymerase epsilon's proofreading function, causing hypermutation (>100-fold increased mutation rate) but not necessarily changing gene expression patterns (Rayner et al., 2016). Both POLE-mutant and wildtype endometrial tumors share similar transcriptional programs for cell growth and hormone signaling, making them difficult to distinguish by mRNA alone (Kandoth et al, 2013). While POLE-mutant tumors have high tumor mutation burden that can trigger immune responses and upregulate immune genes like PD-L1, these immunological signatures are also seen in MSI-high tumors, limiting their specificity for POLE detection (Howitt et al., 2015, Van Gool et al., 2015). The modest classification performance we observed with roughly 88% validation accuracy but poor POLE-mutant recall reflects this biological reality: POLE mutations occur at the DNA level, not the gene expression level. Alternative approaches like mutation signature analysis or TMB quantification from genomic sequencing may provide more direct and accurate POLE detection (Alexandrov et al., 2013). Our findings demonstrate that while feature selection successfully identified genes with statistical associations to POLE status, the severe class imbalance, small minority class size, and disconnect between DNA mutations and mRNA expression limited classifier performance. Future work should prioritize DNA sequencing for POLE detection rather than just mRNA expression alone, as it can be quite noisy and unstable, with more prominent lowly expressed genes that may skew results.

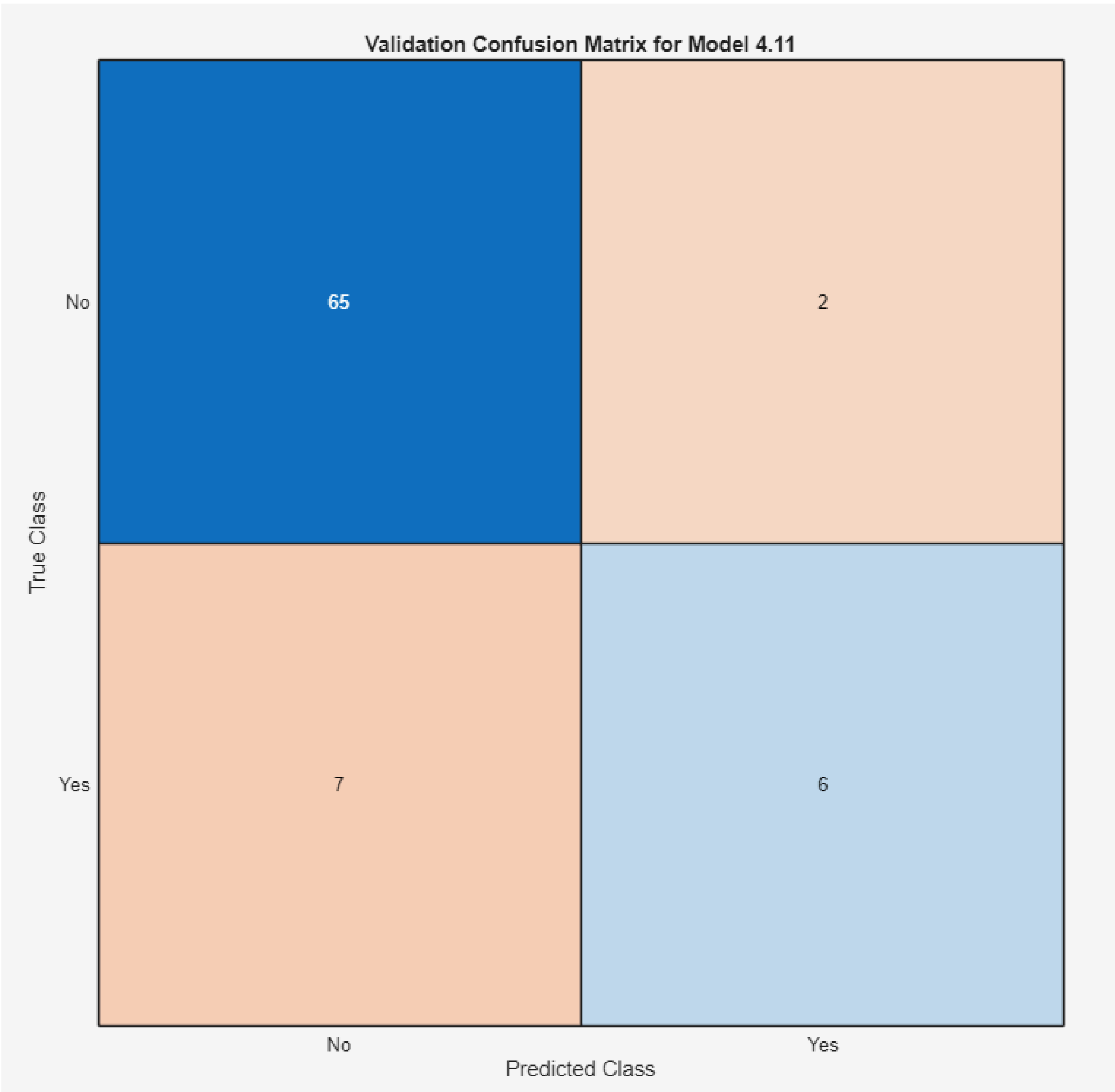


*Figure 37.* Confusion matrix for best performing model, a linear SVM, with 11 features selected and no PCA.

The confusion matrix reveals that while the classifier achieved 88.75% overall accuracy, it correctly identified only 6 of 13 POLE-mutant samples (46.2% recall), misclassifying 7 as wildtype, and incorrectly predicted 2 wildtype samples as mutant, demonstrating modest discriminative ability beyond the 84% baseline achieved by predicting all samples as wildtype.

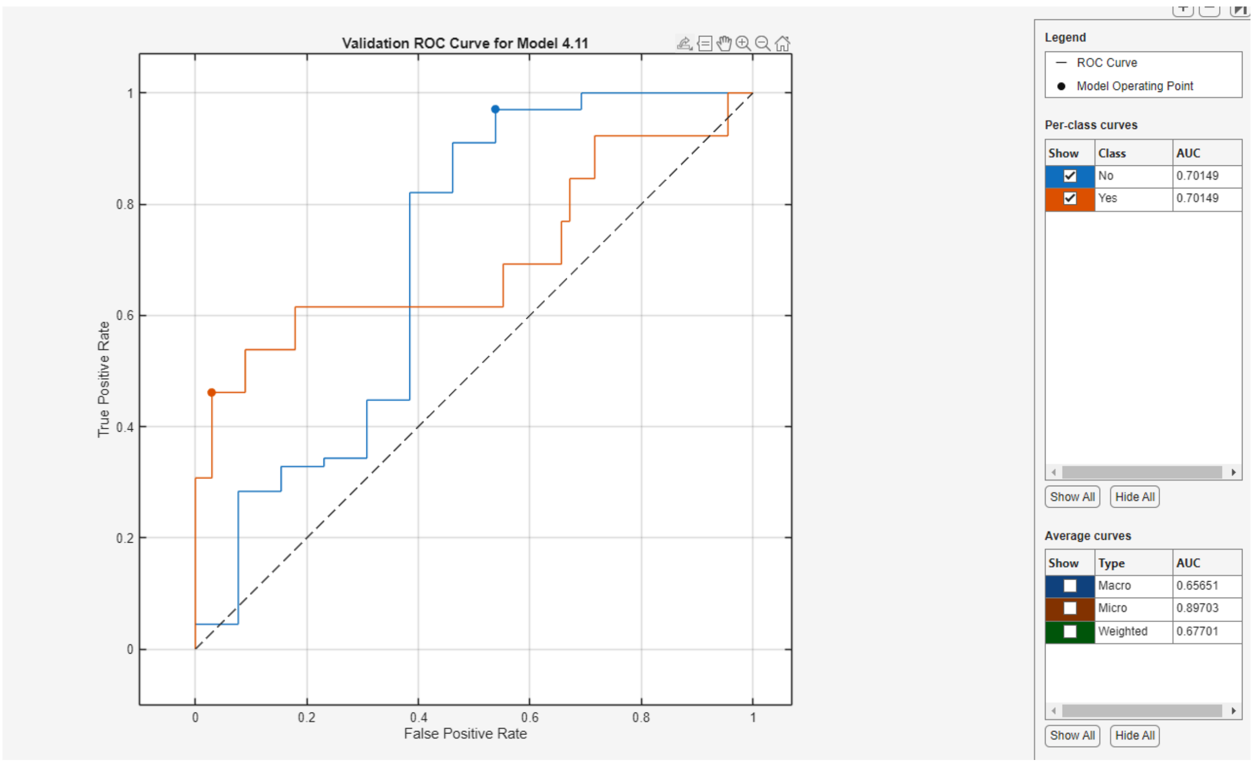


*Figure 38*. ROC Validation curve for the best performing model, a linear SVM, with 11 features selected and no PCA.

The ROC curve shows moderate discrimination ability with AUC values of 0.70149 for the "No" (wildtype) class and 0.70149 for the "Yes" (POLE-mutant) class, indicating performance above random chance where AUC = 0.5 but still far from excellent classification where AUC > 0.9. While the model demonstrates some ability to distinguish POLE-mutant from wildtype samples, the modest AUC values confirm that mRNA expression provides limited discriminative signal for POLE mutation status, or that potential feature selection was poor.

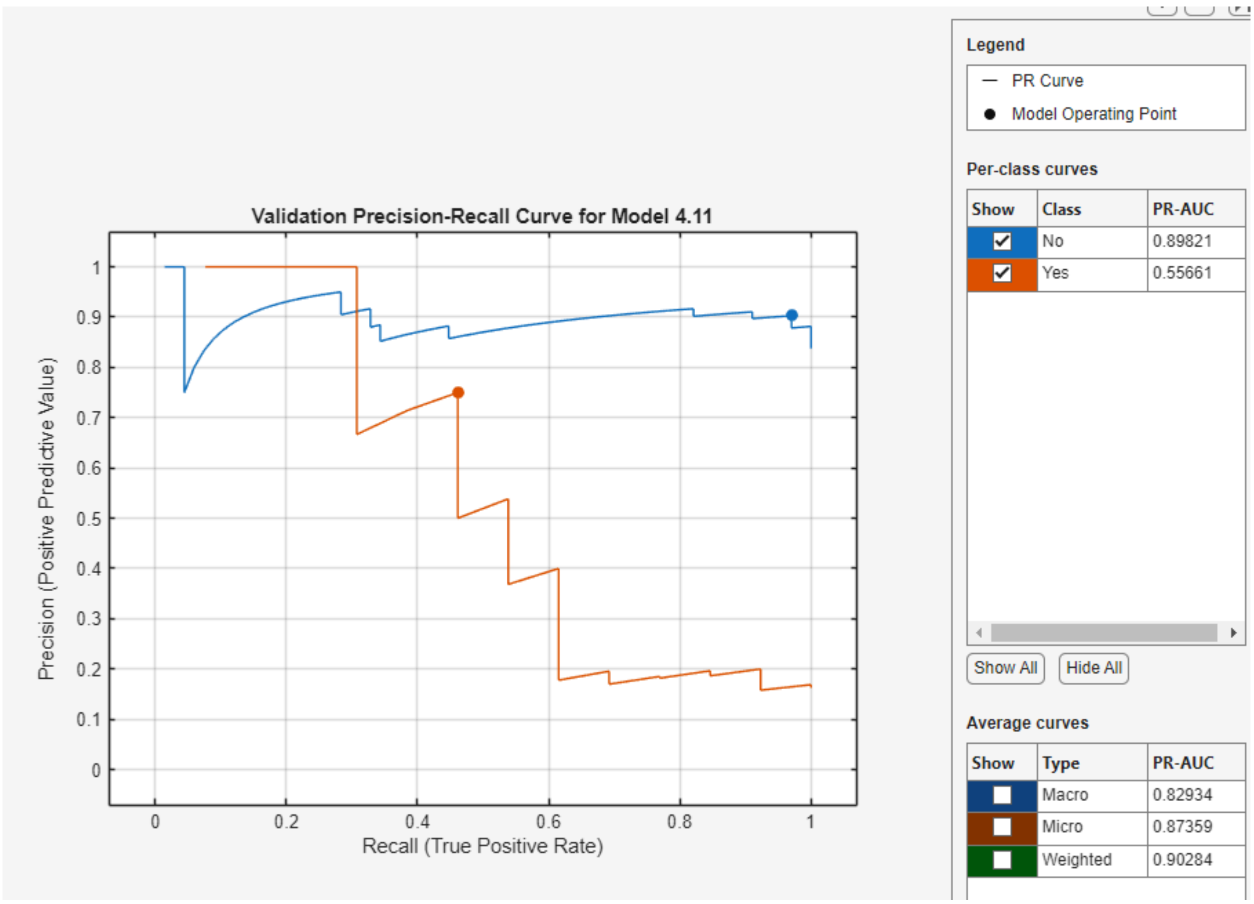


*Figure 39.* Validation precision-recall curve for the best performing model, a linear SVM, with 11 features selected and no PCA.

The precision-recall curves tell two very different stories. While the model excels at identifying wildtype samples (PR-AUC = 0.90), it struggles with POLE-mutant detection (PR-AUC = 0.56). This dramatic gap isn't surprising given the 67:13 class imbalance. It's much easier to correctly predict the abundant wildtype class than to pinpoint rare mutations. The jagged, downward-trending orange curve (POLE-mutant) reveals a trade-off where as the model tries to catch more mutations (higher recall), it becomes less confident in its predictions (lower precision), often misclassifying wildtype samples as mutant. In contrast, the smooth blue curve shows the model can reliably identify wildtype tumors across most thresholds, but this success doesn't translate to the clinically critical task of detecting POLE mutations.

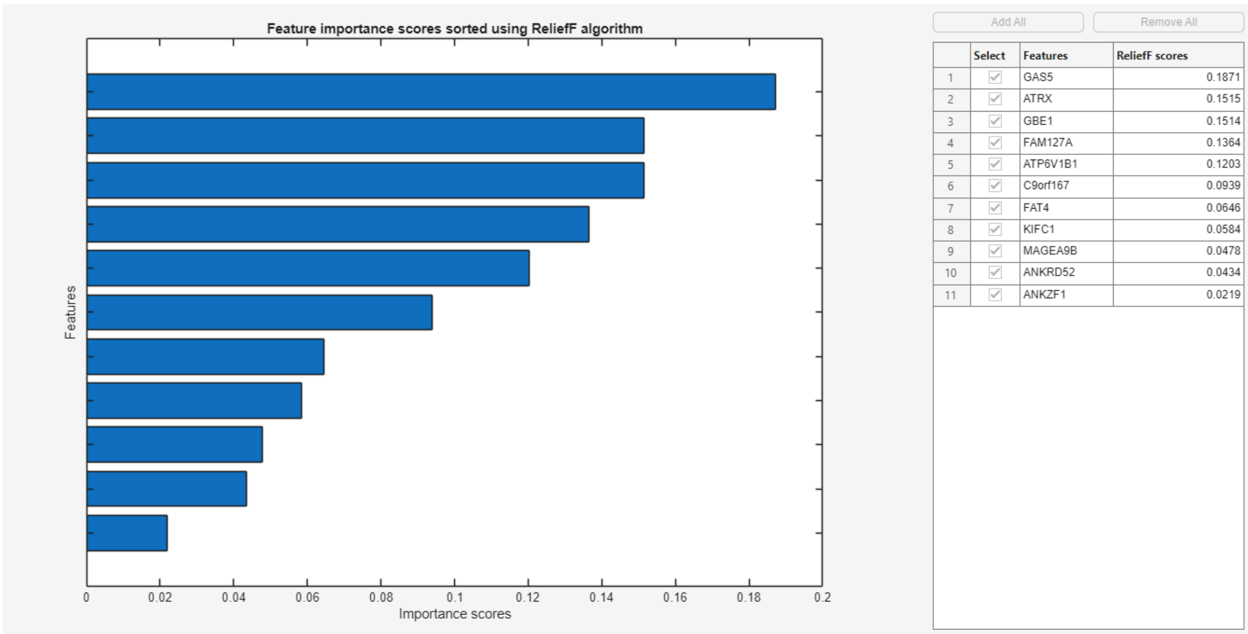


*Figure 40.* Importance Scores using the ReliefF algorithm sorted in descending order.

The feature importance scores from the best-performing SVM model show a steep drop-off in discriminative power across the 11 ReliefF selected genes. GAS5 dominates with a score of 0.1871, followed by ATRX (0.1515) and GBE1 (0.1514), while the bottom four features (FAT4, KIFC1, MAGEA9B, ANKRD52, ANKZF1) all score below 0.065, with ANKZF1 contributing only 0.0219. This suggests that most discriminative power is concentrated in a few top genes, while the remaining features may add noise rather than useful information. Visual inspection of predictor scatterplots confirmed this as nearly all gene pairs showed overlap between POLE-mutant and POLE-wildtype samples with no clear separation, reflecting the challenge that POLE mutations affect DNA replication fidelity, and not solely gene expression patterns. Given this weak contribution from low importance features, removing the bottom 3-4 genes (scores < 0.06) and retaining only the top 7-8 genes with ReliefF scores above 0.10 could potentially reduce overfitting and improve minority class recall by increasing the signal-to-noise ratio. However, even with feature reduction, classification performance may remain limited because mRNA expression simply does not strongly discriminate against POLE mutation status and genomic sequencing approaches rather than expression profiling, although potentially more difficult and costly, are necessary for accurate POLE detection. The feature amount is also already quite low and removing any more potentially valuable information may make differentiating even harder and even worse.

To continue the investigation, algorithms were rerun with PCA enabled where 6/11 features were automatically selected. The best performing models here were a quadratic SVM and a medium Gaussian SVM both with 88.8% accuracy. PCA performs dimensionality reduction by transforming the original 11 correlated gene expression features into a smaller set of uncorrelated principal components that capture the maximum variance in the data, effectively removing redundant information and reducing noise. The automatic selection of 6 components out of 11 suggests that approximately 55% of the feature space contains the majority of the discriminative signal, while the remaining components likely represent noise or redundant variation, aligning with our earlier observation that low importance features (ANKZF1, ANKRD52, MAGEA9B) contribute minimal discriminative power. By retaining only the top 6 principal components, PCA performed automated feature reduction similar to our proposed strategy of keeping high importance genes, but through selecting specific genes weighted by their contribution to the greatest variance.

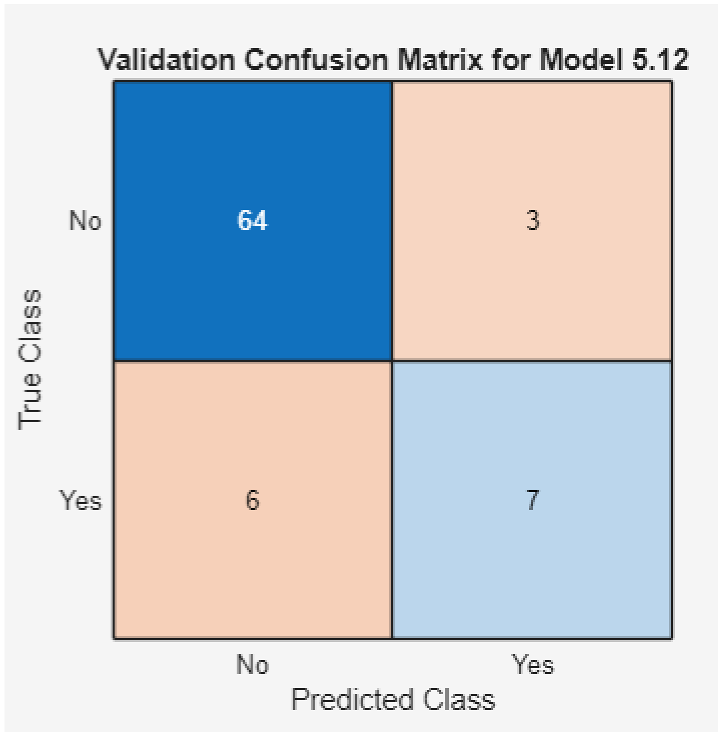


*Figure 41.* Confusion matrix for best performing model, a quadratic SVM, with 6/11 features selected and PCA.

Accuracies between the previous linear SVM and quadratic SVM are essentially the same and so are other metrics below. The quadratic SVM achieved 7 true positive POLE-mutant predictions out of 13 (53.8% recall) compared to the linear SVM's 6 out of 13 (46.2% recall), representing minor improvement in detecting the minority class, catching one additional POLE mutation while maintaining similar false positive rates (3 vs. 2)

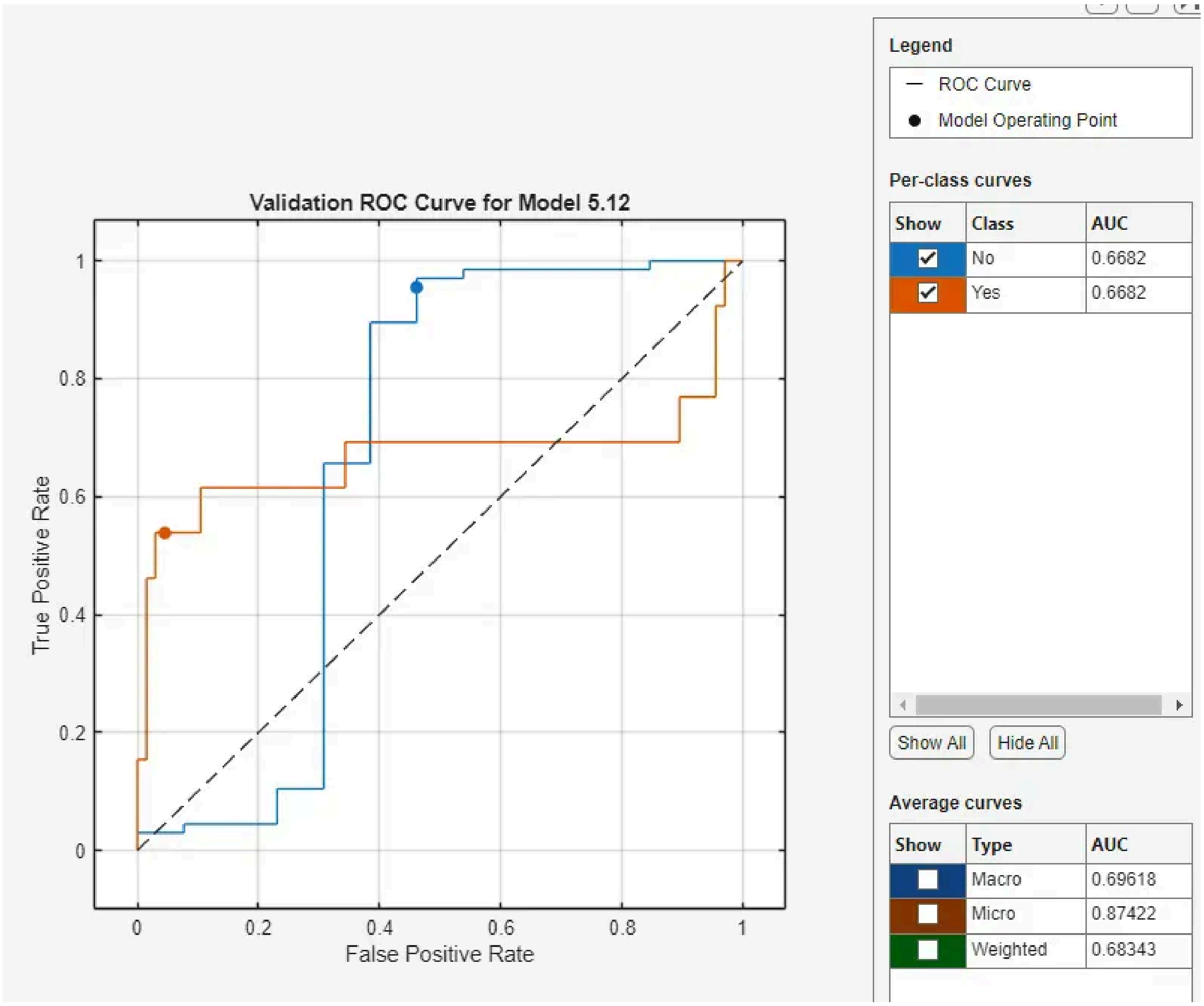


*Figure 42*. ROC Validation curve for the best performing model, a quadratic SVM, with 6/11 features selected and PCA.

Poorer ROC-AUC performance is present here compared to the approximately 0.71 from the linear SVM, with only 0.6682 for both classes here.

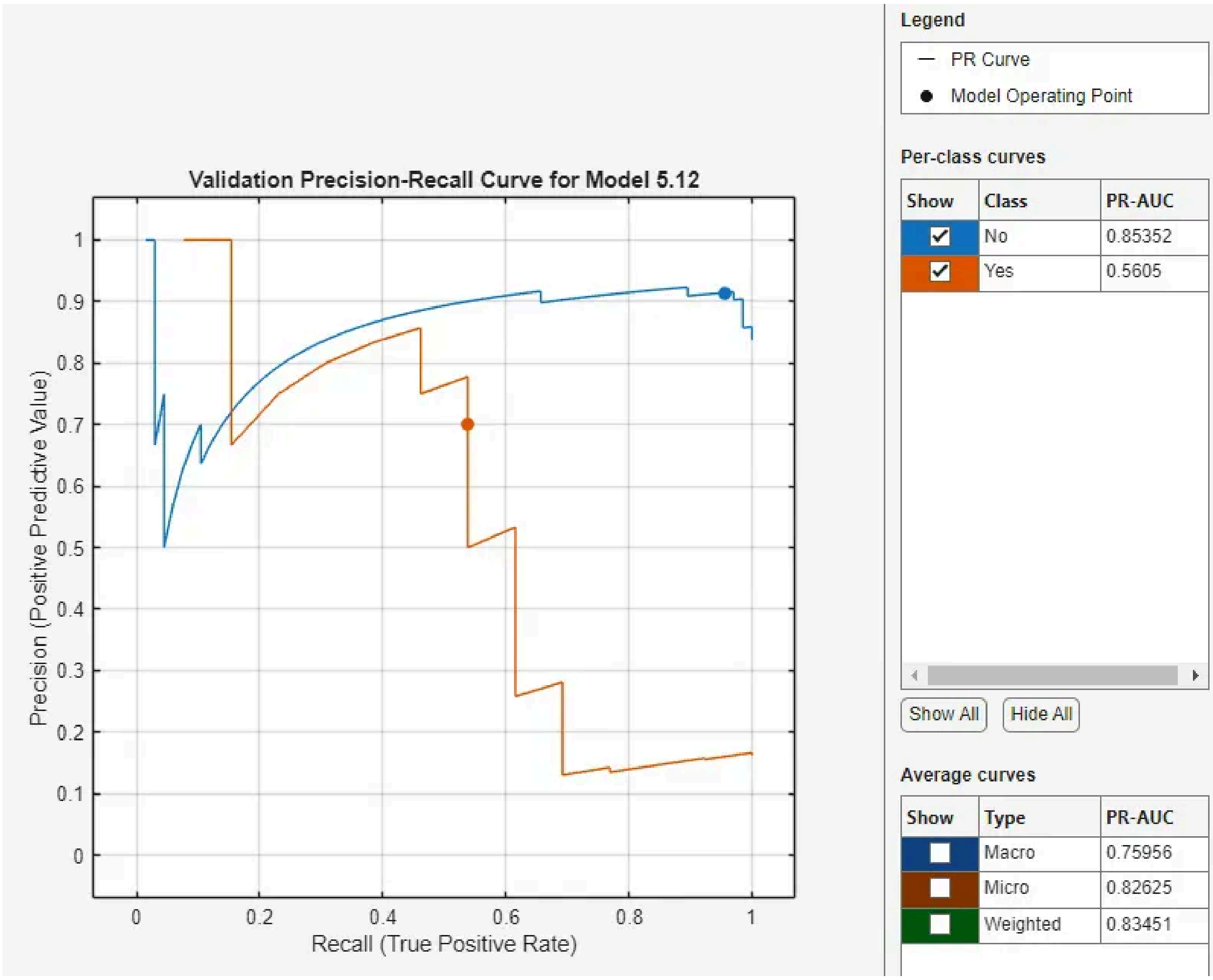


*Figure 43.* Validation precision-recall curve for the best performing model, a quadratic SVM, with 6/11 features selected and PCA.

Again, PR-AUC metrics of 0.85352 for the wildtype class and 0.5605 for the mutant class are essentially the same as the linear SVM. While we received one more true positive overall, the bigger picture is still the same. It looks like approximately 6-8 features (whether selected directly or as principal components) represent the maximum useful information extractable from mRNA expression for this classification task, and further model refinement is unlikely to yield substantial improvements without incorporating alternative data modalities such as genomic sequencing or mutation signatures.

In fact, rerunning the linear SVM with 7/11 features, removing FAT4, ANKZF1, ATP6V1B1, and ANKRD52 produced substantially worse results, with 83.8% accuracy yet catching only 3 true positives instead of the 6/7 above. PR-AUC dropped substantially to 0.39626 for the mutant class and ROC-AUC for both classes hovered around 0.64868. PCA is more robust at finding components of greater variance while manually removing the lowest contributing features is actually obscuring valuable information for our class performance. All in all, however, metrics between the main original linear SVM and quadratic SVM are comparable, but is not a perfect predictor of POLE driver mutation from only the mRNA genes available.

**Question 4**

To explore the relationship between age and the selected mRNAs, we can first look at the levels of measurement for each variable. Both variables, age and mRNA expression are scale measurements. Considering this, Pearson correlation test is the best choice when looking at the association of two scale measures (Morgan et al., 2012).

Assumptions for a Pearson correlation involve that there is a linear relationship between the two variables, outliers are investigated and removed, and the scores of one variable are normally distributed for each value of the other variable (Morgan et al., 2012).

**Correlations**

| | | age | GAS5 | GBE1 | ATRX | KIFC1 | C9orf167 | FAM127A | MAGEA9B | ANKRD52 | ATP6V1B1 | ANKZF1 | FAT4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| age | Pearson Correlation | 1 | -.101 | -.030 | -.049 | .056 | -.050 | .087 | .245* | .051 | .163 | .111 | -.190 |
| | Sig. (2-tailed) | | .324 | .771 | .635 | .584 | .629 | .398 | .016 | .619 | .110 | .278 | .062 |
| | N | 97 | 97 | 97 | 97 | 97 | 97 | 97 | 97 | 97 | 97 | 97 | 97 |
| GAS5 | Pearson Correlation | -.101 | 1 | -.205* | -.121 | -.075 | -.020 | .161 | .055 | -.343** | -.050 | .142 | -.020 |
| | Sig. (2-tailed) | .324 | | .041 | .230 | .456 | .841 | .109 | .588 | <.001 | .622 | .158 | .846 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| GBE1 | Pearson Correlation | -.030 | -.205* | 1 | .514** | -.199* | .097 | -.296** | .199* | .221* | -.069 | -.130 | .198* |
| | Sig. (2-tailed) | .771 | .041 | | <.001 | .047 | .336 | .003 | .048 | .027 | .494 | .196 | .048 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| ATRX | Pearson Correlation | -.049 | -.121 | .514** | 1 | -.229* | .109 | -.314** | .147 | .178 | -.234* | -.270** | .574** |
| | Sig. (2-tailed) | .635 | .230 | <.001 | | .022 | .282 | .001 | .145 | .077 | .019 | .007 | <.001 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| KIFC1 | Pearson Correlation | .056 | -.075 | -.199* | -.229* | 1 | -.145 | -.005 | -.088 | .107 | .144 | .021 | -.235* |
| | Sig. (2-tailed) | .584 | .456 | .047 | .022 | | .150 | .960 | .383 | .290 | .153 | .832 | .019 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| C9orf167 | Pearson Correlation | -.050 | -.020 | .097 | .109 | -.145 | 1 | -.169 | -.027 | .251* | .051 | -.182 | -.161 |
| | Sig. (2-tailed) | .629 | .841 | .336 | .282 | .150 | | .093 | .788 | .012 | .611 | .070 | .111 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| FAM127A | Pearson Correlation | .087 | .161 | -.296** | -.314** | -.005 | -.169 | 1 | .081 | -.168 | .128 | .082 | -.062 |
| | Sig. (2-tailed) | .398 | .109 | .003 | .001 | .960 | .093 | | .425 | .094 | .204 | .417 | .540 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| MAGEA9B | Pearson Correlation | .245* | .055 | .199* | .147 | -.088 | -.027 | .081 | 1 | -.051 | .102 | .144 | .032 |
| | Sig. (2-tailed) | .016 | .588 | .048 | .145 | .383 | .788 | .425 | | .612 | .314 | .152 | .749 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| ANKRD52 | Pearson Correlation | .051 | -.343** | .221* | .178 | .107 | .251* | -.168 | -.051 | 1 | .085 | -.055 | -.020 |
| | Sig. (2-tailed) | .619 | <.001 | .027 | .077 | .290 | .012 | .094 | .612 | | .400 | .587 | .843 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| ATP6V1B1 | Pearson Correlation | .163 | -.050 | -.069 | -.234* | .144 | .051 | .128 | .102 | .085 | 1 | .318** | -.347** |
| | Sig. (2-tailed) | .110 | .622 | .494 | .019 | .153 | .611 | .204 | .314 | .400 | | .001 | <.001 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| ANKZF1 | Pearson Correlation | .111 | .142 | -.130 | -.270** | .021 | -.182 | .082 | .144 | -.055 | .318** | 1 | -.342** |
| | Sig. (2-tailed) | .278 | .158 | .196 | .007 | .832 | .070 | .417 | .152 | .587 | .001 | | <.001 |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| FAT4 | Pearson Correlation | -.190 | -.020 | .198* | .574** | -.235* | -.161 | -.062 | .032 | -.020 | -.347** | -.342** | 1 |
| | Sig. (2-tailed) | .062 | .846 | .048 | <.001 | .019 | .111 | .540 | .749 | .843 | <.001 | <.001 | |
| | N | 97 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Table 2. Pearson Correlation Matrix Between Age and Gene Expression Variables

**Tests of Normality**

| | Kolmogorov-Smirnov[a] | | | Shapiro-Wilk | | |
|---|---|---|---|---|---|---|
| | Statistic | df | Sig. | Statistic | df | Sig. |
| GAS5 | .063 | 97 | .200* | .987 | 97 | .439 |
| GBE1 | .089 | 97 | .054 | .972 | 97 | .036 |
| ATRX | .063 | 97 | .200* | .979 | 97 | .120 |
| KIFC1 | .119 | 97 | .002 | .942 | 97 | <.001 |
| C9orf167 | .118 | 97 | .002 | .941 | 97 | <.001 |
| FAM127A | .071 | 97 | .200* | .984 | 97 | .280 |
| MAGEA9B | .298 | 97 | <.001 | .809 | 97 | <.001 |
| ANKRD52 | .061 | 97 | .200* | .976 | 97 | .074 |
| ATP6V1B1 | .082 | 97 | .107 | .975 | 97 | .066 |
| ANKZF1 | .051 | 97 | .200* | .995 | 97 | .987 |
| FAT4 | .057 | 97 | .200* | .989 | 97 | .625 |
| age | .048 | 97 | .200* | .984 | 97 | .285 |

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 3. Tests of Normality for Age and Gene Expression Variables

Given that the variables of age and miRNA expression are continuous, the Pearson correlation was applied to test the association between age and miRNA expression. Linearity and outliers were checked visually; however, the normality was checked by the Shapiro-Wilk test, which revealed some deviations from normality in some miRNAs as seen in table 3.

Pearson correlation analysis detected a weak but statistically significant positive correlation between age and MAGEA9B expression with r = 0.245 and p = 0.016, and no significant correlations for other miRNAs. In those cases where the assumptions could not be sufficiently fulfilled, Spearman rank correlation would be a non-parametric alternative.

**<u>Question 5</u>**

To identify an appropriate test to look for the association between the combined Stage and POLE driver mutation status, we need to combine the sub stages into a single stage. When combining substages, combined stage 1 included Stage I, IA, IB, and IC. Combined stage 2 included Stage II, and IIB. Combined stage 3 included Stage III, IIIA, IIIB, IIIC, IIIC1, and IIIC2. Combined stage IV included only IVB samples. Each of the stages were represented with roman numerals, either I, II, III, or IV, corresponding to their respective stage number.

Combined stage is an ordinal measure, while POLE driver mutation status is a nominal measure. An appropriate test to explore an association between the two variables is a chi-square. Assumptions for a Chi-square include that the data for variables are independent from one another (Morgan et al., 2012). Data of one sample and their combined stage and POLE driver mutation status does not affect another data of another sample. While combined stage is an ordinal measure, ranging from stage I to stage IV, the Chi-square test treats these measures as nominal (Morgan et al., 2012).

Another assumption of Chi-square tests is that at least 80% of the expected frequencies should be 5 or larger (Morgan et al., 2012). Observed counts are illustrated in the table below.

**nominal * POLE_status Crosstabulation**

| | | | POLE_status | | |
|---|---|---|---|---|---|
| | | | No | Yes | Total |
| nominal | Stage I | Count | 55 | 10 | 65 |
| | | Expected Count | 54.6 | 10.4 | 65.0 |
| | Stage II | Count | 14 | 2 | 16 |
| | | Expected Count | 13.4 | 2.6 | 16.0 |
| | Stage III | Count | 14 | 4 | 18 |
| | | Expected Count | 15.1 | 2.9 | 18.0 |
| | Stage IV | Count | 1 | 0 | 1 |
| | | Expected Count | .8 | .2 | 1.0 |
| Total | | Count | 84 | 16 | 100 |
| | | Expected Count | 84.0 | 16.0 | 100.0 |

*Table 3.* Examining observed counts in combined Stage I, II, III, and IV samples stratified by the presence or absence of POLE driver mutation.

As seen in the table above, there are 4 cells (Combined Stage IV and no mutation, and Combined Stage II, III, and IV and no mutation) that have cells less than a count of 5. This is also observed in the comment illustrated in the table below. As samples were already combined into different stages, a Fisher's exact test can be used instead.

**Chi-Square Tests**

| | Value | df | Asymptotic Significance (2-sided) | Exact Sig. (2-sided) |
|---|---|---|---|---|
| Pearson Chi-Square | .873[a] | 3 | .832 | .817 |
| Likelihood Ratio | .996 | 3 | .802 | .817 |
| Fisher-Freeman-Halton Exact Test | 1.313 | | | .762 |
| N of Valid Cases | 100 | | | |

a. 4 cells (50.0%) have expected count less than 5. The minimum expected count is .16.

*Table 4.* Results of Fisher's Exact Test

As seen in the table above, the results of the Fisher's exact test illustrate there is no significant association between the combined stage of UCEC and if the POLE drive mutation is present or absent ($X^2$ = 1.313, p = 0.762, N = 100).

**Question 6**

a) MSI status

Kaplan-Meier survival analysis was performed to compare overall survival across MSI status groups. Survival time was defined as days to death or last follow-up, with death treated as the event and living patients censored.

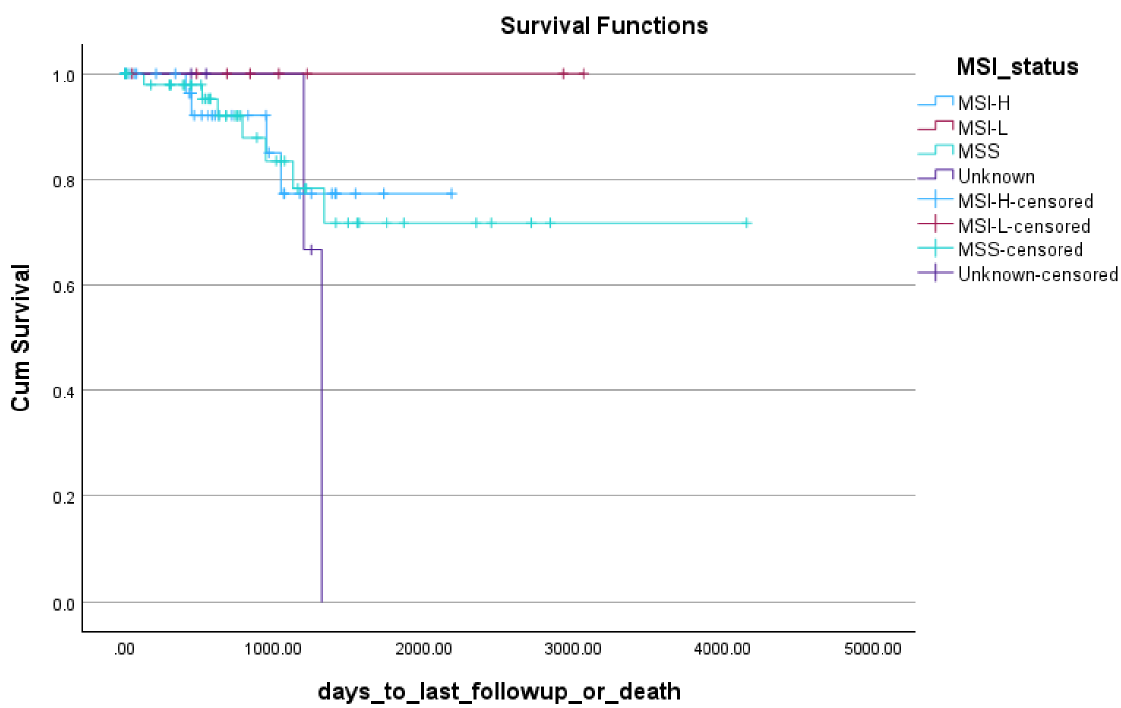

*Figure* 41. Kaplan-Meier Estimates of Overall Survival by MSI status

**Case Processing Summary**

| MSI_status | Total N | N of Events | Censored N | Censored Percent |
|---|---|---|---|---|
| MSI-H | 36 | 4 | 32 | 88.9% |
| MSI-L | 8 | 0 | 8 | 100.0% |
| MSS | 50 | 7 | 43 | 86.0% |
| Unknown | 5 | 2 | 3 | 60.0% |
| Overall | 99 | 13 | 86 | 86.9% |

*Table 5.* Case Processing Summary For Survival Analysis by MSI Status

**Overall Comparisons**

| | Chi-Square | df | Sig. |
|---|---|---|---|
| Log Rank (Mantel-Cox) | 2.841 | 3 | .417 |
| Breslow (Generalized Wilcoxon) | 1.180 | 3 | .758 |
| Tarone-Ware | 1.658 | 3 | .646 |

Test of equality of survival distributions for the different levels of MSI_status.

*Table 6.* Overall Comparison of Survival Curves by MSI Status

Overall survival was assessed with Kaplan-Meier survival methods and compared between molecular subgroups using Log-Rank, Breslow, and Tarone-Ware tests. The results of the Kaplan-Meier curves stratified by MSI status indicated relatively overlapping survival curves among the subgroups with no statistically significant difference in distribution survival functions Figure 41.

This was mirrored by formal testing since the Log-Rank ($\chi^2 = 2.84$, $p = .417$), Breslow ($p = .758$), and Tarone–Ware ($p = .646$) tests did not indicate a difference in survival distribution according to the MSI status Table 6. Median survival time was not reached across MSI status groups as there were only 13 death events that occurred during follow up, with the majority of cases censored as the patients being alive at the last follow up, seen in table 5,.

b) POLE Driver Mutation

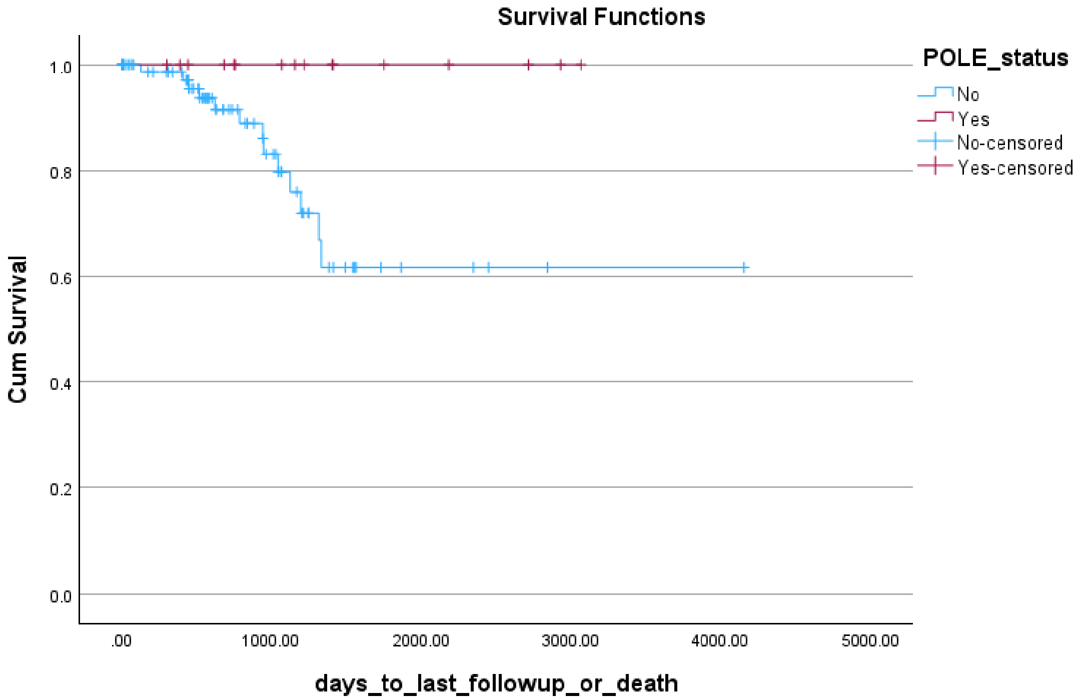


*Figure 42.* Kaplan-Meier overall survival by POLE driver mutation status

**Overall Comparisons**

| | Chi-Square | df | Sig. |
|---|---|---|---|
| Log Rank (Mantel-Cox) | 4.412 | 1 | .036 |
| Breslow (Generalized Wilcoxon) | 3.360 | 1 | .067 |
| Tarone-Ware | 4.030 | 1 | .045 |

Test of equality of survival distributions for the different levels of POLE_status.

Table 7. Case Processing Sum

**Case Processing Summary**

| POLE_status | Total N | N of Events | Censored | |
|---|---|---|---|---|
| | | | N | Percent |
| No | 83 | 13 | 70 | 84.3% |
| Yes | 16 | 0 | 16 | 100.0% |
| Overall | 99 | 13 | 86 | 86.9% |

Table 8. Case Processing Summary table for survival analysis

The overall survival curves were compared between the different status groups of POLE using the Log-Rank, Breslow, and Tarone-Ware tests. The results showed that there is a statistically significant association between the overall survival curves and the different status groups of POLE, with a significant Log-Rank value ($\chi^2 = 4.41$, $p = 0.036$), as well as a Tarone-Ware value ($\chi^2 = 4.03$, $p = 0.045$), but not a Breslow value ($p = 0.067$), indicating that the differences in overall survival are more evident in longer follow-up periods but not in earlier ones (Table 7).

Since no deaths with POLE mutations were observed which can be seen in table 8, there was a complete censoring of patients with POLE mutations. Median survival time was not reached in the POLE-mutated group because all observations were censored and no events were observed during follow-up.

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